



UNIVERSITÀ
DI PAVIA

Dipartimento di Fisica

Corso di laurea in Fisica



UNIVERSITY
OF CAMBRIDGE

Institute of Astronomy

Visiting research project

Measuring pattern speed of spirals in gravitationally unstable protostellar discs

Relazione per la laurea di:

Mauro Ghirardi

Supervisore:

Dr. Cristiano Longarini

Correlatore:

Prof. Andrea Negri

Anno accademico 2024 – 2025

Contents

1	Introduction	3
2	Accretion discs dynamics	5
2.1	Basic equations	5
2.1.1	Centrifugal balance	7
2.1.2	Viscosity and angular momentum transport	8
2.2	The role of self-gravity	10
2.2.1	The development of turbulence	10
2.2.2	Special cases	14
2.2.3	Gravitational instabilities and self-regulation	15
2.2.4	The α -prescription	17
2.3	Spectral energy distribution (SED)	19
3	Protostellar discs	22
3.1	Confirmation of theoretical expectations	22
3.2	Formation, classification and observations	27
3.2.1	Class 0 and I systems	29
3.2.2	Class II and III systems	30
3.3	Planet formation	31
3.3.1	The core accretion model	32
3.3.2	The gravitational instability model	32
3.3.3	Planets and instabilities	33
3.4	Dust properties and dynamics	34
4	Measuring pattern speed	36
4.1	The hydrodynamical simulations and the MCFOST code	37
4.2	Tracking the spiral arms and measuring velocity	39
4.3	Results	43
5	Conclusions	48

CONTENTS

A	Derivation of hydrodynamics equations	50
A.1	Derivative operations in cylindrical coordinates	50
A.2	Vertically integrated continuity equation	51
A.3	Vertically integrated Navier-Stokes equations	51
A.3.1	Vertical component	51
A.3.2	Radial component	53
A.3.3	Azimuthal component	54
B	Generalities on radiative transport	56
B.1	Fundamental definitions	56
B.2	Thermal emission	57
B.3	The Rosseland approximation	59
	Bibliography	61

ABSTRACT

Accretion discs dynamics is governed by a phenomenon akin to viscosity determining an outward flow of angular momentum that allows accretion and whose deep origin is still unknown. Nowadays, a great interest towards discs' self-gravity has developed since it has been demonstrated that it can play a fundamental role in the growth of instabilities leading to a sustained transport of angular momentum even in very light discs.

Numerical simulations have almost completely confirmed the theoretical predictions about the evolution of gravitationally unstable discs. However, it's still not clearly evident whether gravitational instabilities can be distinguished from other kinds of instabilities (e.g. magneto-rotational instabilities or planetary spirals) from direct observations.

Given this uncertainty, the ability to perform precise measurements of quantities comparable to the results of numerical simulations becomes crucial. In this thesis, in particular, the discussion will focus on the direct measurement of the pattern angular velocity of the spiral structures created by gravitational instabilities in protostellar discs.

From the analysis of simulated observations of the Atacama Large Millimeter Array of discs with different masses, we will find that on a 10-year timescale the pattern velocity can indeed be measured. The obtained trend will generally be in tune with a Keplerian profile, with a measured speed slightly exceeding the pure Keplerian angular velocity of the system. This result will be, moreover, consistent with the same analysis carried out directly on snapshots of hydrodynamical simulations.

Chapter 1

Introduction

An accretion disc is an astrophysical system in which a central object is fed by the surrounding rotating material due to viscous forces. In particular, in this thesis, we will focus on discs around *Young Stellar Objects (YSO)* like a protostar. Such systems exhibit extreme variability in their properties, which can also change a lot within different parts of the same disc.

A primary phenomenon occurring in these structures is the redistribution of angular momentum, a process needed to allow accretion whose cause is still not fully understood. Indeed, classical collisional viscosity does not provide enough torque to justify the magnitude and timescales of this evolution, so alternative ways are being sought.

A possible solution to the problem can come from the effects of the disc's self-gravity, in particular from gravitational instabilities, which develop into characteristic spiral structures in the disc's density. Such a phenomenon is usually described adopting the so-called α -prescription, which allows us to model the transport as due to an "effective viscosity"; however, we'll highlight some limits of this approach.

One of the characteristic features of gravitational instabilities is their pattern speed, which is expected to scale like the Keplerian angular velocity. In this work, the capability to measure this velocity from direct observations will be discussed; in practice, this matter will be accomplished by studying simulated data produced using the radiative transfer code MCFOST.

Throughout the whole work, the essentiality of numerical simulations to study these arrangements will also emerge.

The structure of the thesis is as follows. In Chapter 2 the fundamental equations regulating the dynamics of accretion discs will be exposed, showing what can and, more importantly, what cannot be justified and interpreted up to our knowledge. In Chapter 3 we'll then turn our attention to protostellar discs specifically, presenting some results of numerical simulations which validate many theoretical previsions.

CHAPTER 1. INTRODUCTION

Moreover, we'll describe the formation and classification of these systems and we'll investigate the matter of planet formation. Finally, in Chapter 4, we'll move to a deeper analysis of the issue of measuring the pattern speed of perturbations in a gravitationally unstable disc, presenting the main features of the code used for the data processing and the results obtained.

For the sake of completeness, in Appendix A a more detailed derivation of some of the most fundamental equations presented in Chapter 2 is reported, while in Appendix B some generalities about radiative transport are recalled.

Chapter 2

Accretion discs dynamics

2.1 Basic equations

Accretion discs can usually be studied in the *thin disc approximation*, considering the thickness H of the disc to be much smaller than the radial distance from the central object R . In protostellar discs, for example, $H/R \approx 0.1$.

Treating the disc as infinitesimally thin will allow us to integrate almost all the equations in the vertical direction, hence dealing with quantities per unit surface or vertically averaged ones¹. For instance, we will largely use the surface density $\Sigma = \int_{-\infty}^{\infty} \rho dz$. We will moreover consider the disc as vertically confined, neglecting the vertical component of the velocity.

A first approach in the description of these systems is employing the classical equations of hydrodynamics, given that the dynamical active element is typically gaseous. Protostellar discs, for example, are made up of gas for about 99% of their mass, with the residual part accomplished by dust which, however, plays a fundamental role in the emission properties of the system, as pointed out in the next chapter.

Such equations are the Navier-Stokes ones and the continuity equation:

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} \right) = -\nabla P + \nabla \cdot \boldsymbol{\sigma} - \rho \nabla \Phi , \quad (2.1)$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 , \quad (2.2)$$

where $\mathbf{v}$ is the velocity of the fluid, ρ is its density, P is the pressure, Φ is the gravitational potential, and $\boldsymbol{\sigma}$ is the stress tensor describing viscosity.

¹Especially when dealing with these kind of quantities, particular caution must be taken to verify the applicability of the thin disc approximation

CHAPTER 2. ACCRETION DISCS DYNAMICS

Eq. 2.1 is simply an expression of Newton's II Law, seeing that the left-hand side represents the acceleration, including the convection term due to velocity gradients², while the right-hand side is made up of the different forces acting on the fluid.

The gravitational potential is defined as the solution of the Poisson equation:

$$\Delta\Phi = 4\pi G\rho(\mathbf{r}) \quad (2.3)$$

where G is the gravitational constant. This equation directly descends from Gauss' theorem, and its solution strongly depends on the density distribution; for our purposes, analytical solutions can be found employing some approximations.

The stress tensor needed for accretion should be a pure shear one, so that its components are proportional to velocity gradients and the dominant component is the $R\varphi$ one. The proportionality factor η depends on the nature of the viscosity in the disc which is, however, still unknown. Since it allows us to derive some important general results, we'll start by considering a classical collisional viscosity, and we'll show the limits of this approach.

These equations can be rewritten using cylindrical polar coordinates due to the symmetry of the problem, so they can be broken down into the three components; each one can be studied in the limit cases in which gravity is dominated by the central object (*Keplerian rotation*) or by the disc's self-gravity (*self-gravitating disc*); in some situations, an analytical solution can be found.

For instance, employing the approximations described in the next section for the radial equation, we can first solve the vertical hydrostatic equilibrium, find an analytical solution for $\rho(z)$ in both limits³, and define the typical scales for the disc thickness in the non-self-gravitating regime and in the opposite one:

$$H_{nsg} \doteq \frac{c_s}{v_\varphi} R \quad \text{and} \quad H_{sg} \doteq \frac{c_s^2}{\pi G \Sigma}, \quad (2.4)$$

where c_s is the speed of sound and v_φ is the azimuthal component of the velocity. In a realistic system, however, both the contributions to the gravitational potential are present, so, balancing their action, it is possible to determine when self-gravity becomes dominating: we find out that the vertical structure of the disc is affected by self-gravity even when the disc mass is much lower than the central object's mass, as we obtain $\frac{M_D}{M_*} \approx \frac{H}{R}$, which is a small quantity in the thin disc approximation. By contrast, self-gravity has an impact on the radial structure of the disc only if the two

²The whole expression is called *Lagrangian derivative* of the velocity; this derivative is the "natural" one in the Lagrangian approach to hydrodynamics, which studies the evolution of *fluid elements* rather than regarding the physical quantities as fixed in space (*Eulerian approach*).

³Take a look at Appendix A or consult [Lodato, 2007] for a detailed derivation.

CHAPTER 2. ACCRETION DISCS DYNAMICS

masses are approximately the same, for the star gravity is much more important in this instance⁴.

Moreover, it's important to notice that, from the definition of H_{nsg} given above, a natural ordering between velocities emerges as an equivalent condition to the thin disc approximation, namely $c_s \ll v_\varphi$.

In the next sections, we will focus a bit more on the radial and azimuthal equations. The first one is important to discuss the centrifugal equilibrium and derive the rotation curve of the disc ($v_\varphi(R)$), which will play a crucial role in planet formation, as we will see in the next chapter (3.3.1); from the second one, instead, the equation regulating the transport of angular momentum and the relation $v_R \ll c_s$ (v_R is the radial component of the velocity) can be derived.

More details about the mathematical derivation are given in Appendix A.

2.1.1 Centrifugal balance

The radial component of Eq. 2.1 can be simplified assuming the disc is axisymmetric, so that the derivatives with respect to the azimuthal angle vanish⁵, considering a classical shear viscosity, and adopting the ordering of velocities established above. We are therefore left with

$$\frac{v_\varphi^2}{R} = \frac{1}{\rho} \frac{\partial P}{\partial R} + \frac{\partial \Phi}{\partial R}. \quad (2.5)$$

Considering a barotropic⁶ gas, $c_s^2 \doteq \frac{dP}{d\rho}$ (this comes straightforward from the wave equation for sound waves, which are classically understood as density or pressure waves in a fluid), so the first term on the right-hand side is in first approximation negligible and we can simply derive the rotation curve of the gas in the two limits mentioned above: for a Keplerian system, the rotation curve is the classical $v_\varphi(R) = \sqrt{\frac{GM}{R}}$, while for self-gravitating discs it will depend on the model; if we consider the Mestel model, where the disc extends to infinity and the density distribution has the form $\Sigma(R) = \Sigma_0 \frac{R_0}{R}$, $v_\varphi = \sqrt{2\pi G \Sigma_0 R_0} = \text{const.}$

Since the total disc mass increases with the distance from the central body, realistic models generally include both the contributions from the star and the disc gravities, and a radius at which the transition between the two regimes occurs can be, therefore, determined. Since the material of the disc gradually accretes onto the star, the importance of the disc decreases over time, and the radius of transition moves further from

⁴Once again, refer to App. A or [Lodato, 2007].

⁵In this derivation, as for Eq. 2.7 and 2.10 later, it is necessary to express correctly the derivative operators acting on vector in cylindrical coordinates; check [Adeeb, 2020] for the details.

⁶A fluid is said to be barotropic if its state equation can be written as $P = P(\rho)$, namely the pressure can be expressed solely in terms of the density.

CHAPTER 2. ACCRETION DISCS DYNAMICS

the star. An estimation of the transition radius therefore needs to take into account the accretion rate $\dot{M}_D(R)$, the mass flowing towards the star across a cylindrical surface of radius R , which is simply given by

$$\dot{M}_D(R) = - \int_{\partial\Omega} \rho \mathbf{v} \cdot \mathbf{dA} = - \int_0^{2\pi} \int_{-\infty}^{\infty} \rho v_R (R dz d\varphi) = -2\pi R \tilde{v}_R \Sigma(R) , \quad (2.6)$$

where $\tilde{v}_R$ is the vertically averaged radial velocity and we've neglected the vertical flux since $v_z \sim 0$.

It's important to notice that in its most general expression, the rotation curve can be given in terms of the accretion rate. From this relation, it turns out that a very large growth rate is needed for a disc to be significantly non-Keplerian (i.e., have a rotation curve that strongly differs from the Keplerian one) at finite radii, which is typically not the case for protostellar discs.

Despite their low magnitude, pressure gradients play a very important role when considering the interaction between the gas and solid components in the disc: the latter are not affected by pressure and will hence describe precise Keplerian orbits (neglecting for the moment the disc self-gravity) with a slightly different velocity with respect to the gas (that will be typically sub-Keplerian since the density normally decreases with the radius). This will cause a radial motion of the solid particles, with important consequences described later in sect. 3.3.1.

Moreover, by definition, c_s is a function of the temperature of the disc; more precisely $c_s \sim T^{1/2}$, and it can be demonstrated that in both the cases of active and passive discs⁷, $T \sim R^{-p}$, where $1/2 \leq p \leq 3/4$ depending on the geometry and optical properties of the disc. Therefore, for a Keplerian disc,

$$H = \frac{c_s}{\Omega_k} \sim \frac{R^{-p/2}}{R^{-3/2}} \sim R^{(3-p)/2} ,$$

so the ratio H/R increases with the radius (as $(3-p)/2 > 1$) leading to sizable deviations from the Keplerian behaviour in the outermost regions of the disc (cf. Eq. A.16).

2.1.2 Viscosity and angular momentum transport

Let's start this analysis by integrating Eq. 2.2 in the vertical direction; under the same assumptions as in the previous section, we obtain

$$\frac{\partial \Sigma}{\partial t} + \frac{1}{R} \frac{\partial}{\partial R} (R \Sigma \tilde{v}_R) = 0 \quad (2.7)$$

⁷We'll present this distinction later in sect. 2.3. See [Armitage, 2020] for the derivation of the result used above.

CHAPTER 2. ACCRETION DISCS DYNAMICS

where $\tilde{v}_R$ is again the vertically averaged radial velocity⁸.

Considering now a stress tensor describing the classical shear viscosity⁹, the only non-vanishing entry is $\sigma_{R\varphi} = \eta R \frac{\partial \Omega}{\partial R}$ (η is the shear viscosity coefficient and $\Omega = \frac{v_\varphi}{R}$ is the angular velocity). We can then define the vertically averaged kinematic viscosity ν as

$$\nu = \frac{1}{\Sigma} \int_{-\infty}^{\infty} \eta dz, \quad (2.8)$$

and integrate the φ component of Eq. 2.1 in the vertical direction (later on, we'll denote with $T_{R\varphi}$ the vertical integral of the stress tensor). The terms involving derivatives in the azimuthal direction vanish (like in Eq. 2.7), so we are left with¹⁰

$$\Sigma \left(\frac{\partial v_\varphi}{\partial t} + v_R \frac{\partial v_\varphi}{\partial R} \right) = \frac{1}{R} \frac{\partial}{\partial R} \left(\nu \Sigma R^2 \frac{\partial \Omega}{\partial R} \right). \quad (2.9)$$

Combining this with Eq. 2.7 (cf. App. A.3.3), we recover the equation regulating angular momentum (per unit area) conservation, namely

$$\frac{\partial}{\partial t} (\Sigma R^2 \Omega) + \frac{1}{R} \frac{\partial}{\partial R} (\Sigma R^3 \Omega v_R) = \frac{1}{R} \frac{\partial}{\partial R} \left(\nu \Sigma R^3 \frac{\partial \Omega}{\partial R} \right), \quad (2.10)$$

where the second term on the left-hand side is the radial transport of angular momentum (negative if angular momentum is flowing in), and the right-hand side expresses the action of the torque $G(R) = 2\pi R^2 T_{R\varphi}$ of viscous forces. Let's highlight that both these terms are divergences, thus they describe transport, not dissipation¹¹.

It is important to emphasize the dependency of this flux on the derivative of the angular velocity¹², since it can justify the outward flow of angular momentum needed to allow accretion according to the process described in the next chapter.

Using the continuity equation again, we obtain the expression for the radial velocity and, substituting it back in Eq. 2.7 we find the master equation describing the evolution of accretion discs:

$$\frac{\partial \Sigma}{\partial t} = -\frac{1}{R} \frac{\partial}{\partial R} \left[\frac{1}{\frac{\partial}{\partial R} (R^2 \Omega)} \frac{\partial}{\partial R} \left(\nu \Sigma R^3 \frac{\partial \Omega}{\partial R} \right) \right]. \quad (2.11)$$

⁸In the following equations the $\sim$ symbol will be omitted.

⁹Viscous forces depend on velocity gradients. If gradients are orthogonal to the direction of motion we talk about *shear viscosity*; if they are parallel, we deal with *bulk viscosity*. Let's moreover specify that we'll deal with *Newtonian fluids* only, namely fluids in which the relation between the stress tensor and velocity gradients is linear.

¹⁰We assume the fields involved are sufficiently well-behaving so that $\int_{-\infty}^{\infty} \frac{\partial f}{\partial z} dz = 0$

¹¹When considering the integral form of an equation, the divergence terms can be converted into surface integrals through Gauss theorem, thus they only account for the variation of the given quantity on the borders of the region considered, which is ultimately due to inwards or outwards transport. Dissipation (or generation) would be instead described by volume terms.

¹²When trying to explain this phenomenon in kinetics terms, a paradox seem to arise, but refer to [Lodato, 2008] sect. 4.5.1 for a brief resolution.

CHAPTER 2. ACCRETION DISCS DYNAMICS

In the particular case of ν and Ω independent from Σ , this reduces to a simple diffusion equation for the surface density Σ , whose evolution is ergo determined by viscosity, taking place on a timescale of the order of $\tau_\nu = R^2/\nu$. As evident, this is a local timescale, as it is different for each annulus in the disc depending on the radius (this is consistent with the fact that viscosity is a local phenomenon), but it can also be used to derive a global timescale for the evolution of the disc, considering the outermost annulus (that is associated with the longest timescale).

Assuming viscosity to be originated by collisions between gas molecules, we can derive an expression for ν and therefore compare this timescale to the one associated with the dynamics of the disc $\tau_d = \frac{T_{orb}}{2\pi} = \Omega^{-1}$.¹³ However, if we consider protostellar discs, this argument leads to timescales longer than the age of the Universe, as pointed out in [Lodato, 2007] (sect. 2.5).

On the other hand, the ratio between the two timescales is equal to the Reynolds number $R_e = \frac{Rv_\varphi}{\nu} = \frac{R^2\Omega}{\nu}$; since this number is generally very high, as just said, the fluid is likely to be in a turbulent condition. This suggests ascribing the transport of angular momentum to the mixing of fluid elements moving because of turbulence, rather than to classical viscosity. Unfortunately, we are not able to describe this process from first principles yet, but many hypotheses, including magnetohydrodynamic or gravitational instabilities acting like viscous forces, have been proposed.

2.2 The role of self-gravity

2.2.1 The development of turbulence

A turbulence is defined as a motion characterized by chaotic changes in pressure and fluid velocity leading to vortex formation. In the NS equations, the generation of turbulence is described by the convection term, which exactly describes the acceleration due to rotational vorticose motion, while the viscous term is responsible for its dissipation; interestingly, it can be, however, demonstrated that without viscous-like forces turbulence cannot originate.

The first issue to be tackled when trying to justify the presence of turbulence is finding a mechanism capable of extracting rotational energy to keep the turbulence active. This analysis is accomplished using the standard perturbation theory, which explains how a perturbation, described as a density wave of the form $e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})}$ (where ω is the angular frequency of the perturbation) evolves under the effect of various forces.

In the thin disc approximation, we can express the scalar product in cylindrical coordinates neglecting the vertical component, so the wave takes the form $e^{i(\omega t - kR - m\varphi)}$,

¹³It's interesting to notice that this is also the typical timescale needed to reach hydrostatic equilibrium in the vertical direction for a Keplerian system. See [Lodato, 2008] sect. 6 for the details.

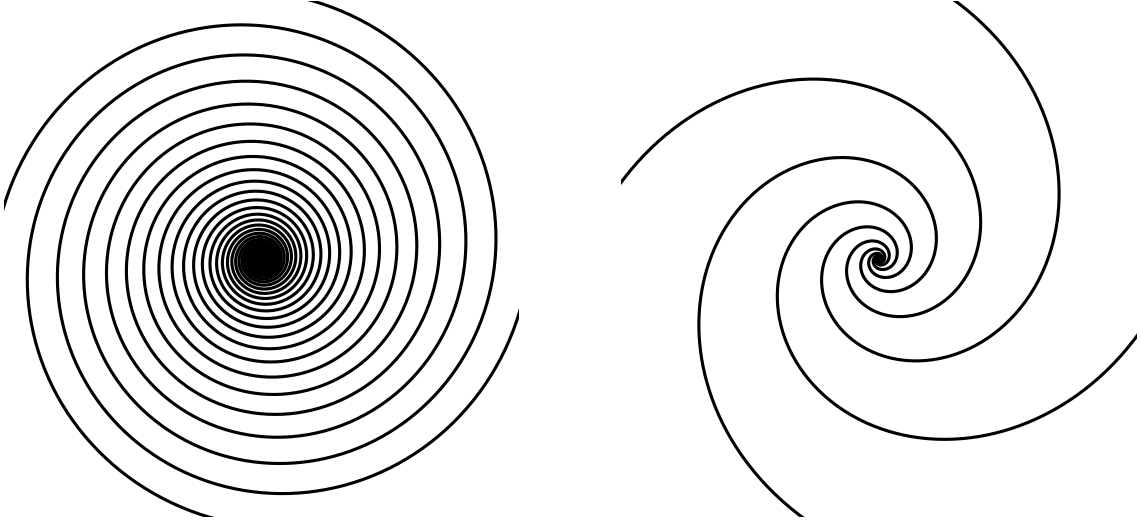


Figure 2.1: Spiral structures with $m = 4$ and pitch angles $\psi = 5^\circ$ (left-hand panel) and 20° (right-hand panel). If the rotation is anticlockwise, these are leading structures.

where k is the radial wavenumber and m is the azimuthal wavenumber. Let's highlight that, since the system must have an angular periodicity of 2π radians, m can only assume integer values, and this value corresponds to the number of angular maxima of the perturbation; in other words, it represents the number of arms of the spiral structure (it can be demonstrated that perturbations evolve in a way that precisely leads to the formation of spiral structures in the disc density).

Let's start from Eq. 2.1, focusing in particular on the case of a self-gravitating disc: having established that classical viscosity has a very small influence on the disc dynamics, we can neglect the viscosity term (obtaining therefore the simple Euler equation) and carry out a linear local analysis of the behaviour of density waves in the WKB approximation¹⁴, which in this context corresponds to dealing with a tightly wound perturbation, where the pitch angle ψ is such that $\tan \psi = \frac{m}{kR} \ll 1$. Combining the linearized continuity, Euler, and Poisson equations for an infinitesimally thin self-gravitating disc, this dispersion relation is found:

$$(\omega - m\Omega)^2 = c_s^2 k^2 - 2\pi G \Sigma |k| + \kappa^2, \quad (2.12)$$

¹⁴The interpretation of this approximation is that the typical scale for the variation of the background properties of the system is much longer than the wavelength of the perturbation, so that it can be locally treated as a plain wave.

CHAPTER 2. ACCRETION DISCS DYNAMICS

where κ is the epicyclic frequency¹⁵, given by

$$\kappa^2 = \frac{2\Omega}{R} \frac{d(\Omega R^2)}{dR} . \quad (2.13)$$

The left-hand side represents the Doppler-shifted frequency, which is the outcome of a balance between pressure, self-gravity, and rotation. It is evident that self-gravity is the term that makes the system unstable, since it is negative: if it dominates, the frequency acquires an imaginary part which results in an exponentially growing perturbation.

Depending on the signs of k , these spiral structures can be trailing ($k > 0$, the spiral arms curve in the direction of rotation) or leading ($k < 0$, the spiral arms bend opposite to the direction of rotation). Once the perturbation is launched, the forcing that generates it typically extinguishes, so that the orientation of the spiral can be modified by the shear between adjacent regions of the disc; hence, the sign of $d\Omega/dR$ becomes the discriminating factor. In protostellar discs, where it is usually negative due to the quasi-Keplerianity of these systems, leading waves are converted into trailing ones within a few orbital periods, so the first ones are hardly ever observed.

Another important feature of such waves is the location where they originate: to produce a perturbation, an exchange of energy and angular momentum between the disc and the perturbing potential is needed, and this can in principle occur everywhere, especially in the case of self-gravitating discs or strong transient perturbations. There are, however, some special spots where this transfer is very efficient due to resonances; for instance, if the pattern speed of the wave $\Omega_p(R) \doteq \omega/m$ is equal to $\Omega(R)$, so that the wave pattern rotates at the same speed as the disc (co-rotation), the perturbation stays in phase with the orbit, thus the torque that it exerts piles up over various periods, determining a sustained transfer of angular momentum. On the other hand, when a difference between the velocities is present, the torque averages out over time, resulting in just a little net transfer of angular momentum.

In addition, let's keep in mind that, in their evolution, perturbations generally spread both inwards and outwards within the disc, but quantities such as energy and angular momentum will flow in a precise direction given by the group velocity $v_g \doteq \frac{\partial \omega}{\partial k}$, which gives the direction in which wave packets propagate.

Finally, it's important to notice that, even though the linear approach can be adopted at first approximation, the evolution of these waves is generally non-linear for several factors, unless they are very weak; thus, the spiral structures will almost always consist of shock waves.

Let's now briefly recall some fundamental features of shocks, as they'll be meaningful later on.

¹⁵This is the natural frequency of small radial oscillations of a fluid element around a circular orbit, originating from the balance of gravity and centrifugal forces acting on a fluid element which is slightly displaced from its natural orbit.

CHAPTER 2. ACCRETION DISCS DYNAMICS

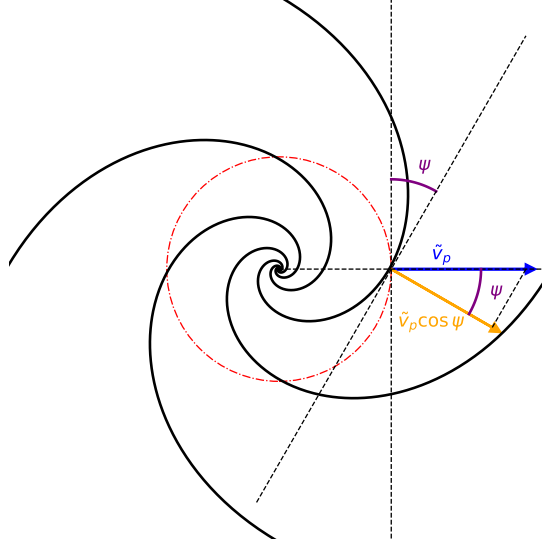


Figure 2.2: Logarithmic spiral with pitch angle $\psi = 30^\circ$. The normal direction to the shock wavefront is shown too.

A shock can be characterised as a thin region where the properties of a fluid change abruptly. A shock front can form and be sustained only if the perturbative wave is supersonic in the direction normal to the front. More precisely, what matters is the velocity in the medium reference frame so, in the case of a rotating disc, a shock forms if the relative velocity between the gas flow and the perturbation is greater than the speed of sound. To handle this properly, we should then consider both the radial and angular components of the velocities but, since we can define the radial phase speed of the waves simply as

$$v_p \doteq \frac{dR}{dt} = \frac{\partial X / \partial t}{\partial X / \partial R} = -\frac{\omega}{k} = -\frac{m\Omega}{k}, \quad (2.14)$$

where $X = kR + m\varphi - \omega t$, we can then define the Doppler-shifted radial phase speed $\tilde{v}_p \doteq \frac{m|\Omega_p - \Omega|}{k}$, which accounts for the difference in rotation as it is the radial velocity in the rotating flow reference system; we can now simply compare this to the radial velocity of the flow v_R which is, however, much smaller than c_s , so we can neglect it when evaluating the difference. Moreover, we have to consider that the direction normal to the shock is not the radial one, but the normal to the spiral arm, since the shock wavefront precisely defines the spirals (Fig. 2.2); all this considered, we are left with the condition

$$\tilde{v}_p \cos \psi \gtrsim c_s, \quad (2.15)$$

which can also be re-expressed in terms of the Doppler-shifted phase Mach number

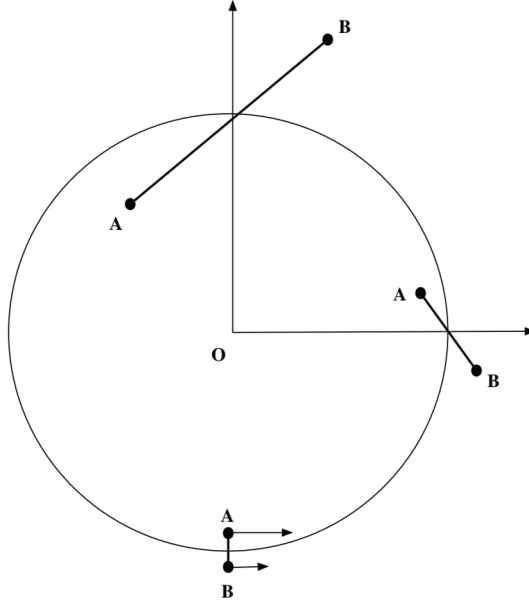


Figure 2.3: Sketch of the process leading to magneto-rotational instabilities (MRI). Figure from [Lodato, 2008].

$\tilde{\mathcal{M}} \doteq \frac{\tilde{v}_p}{c_s}$ as

$$\tilde{\mathcal{M}} \gtrsim \frac{1}{\cos \psi} . \quad (2.16)$$

Bearing in mind that in the tightly wound approximation $\tan \psi \ll 1 \Rightarrow \cos \psi \approx 1$, we expect this number to be very close to unity in such systems.

2.2.2 Special cases

We will focus on the impact of self-gravity in the next section, but first let's make some short considerations about some special cases.

If we perform the derivation described above in the simple case of a purely hydrodynamical system, hence neglecting the pressure and gravity terms, the right-hand side of Eq. 2.12 is simply given by κ^2 , which is typically non-negative for protostellar discs, given that they are only marginally non-Keplerian (as observed in sect. 2.1.1). As a consequence, purely hydrodynamical discs are generally linearly stable and cannot develop turbulence.

A low magnetic field could, in fact, be a source of turbulence. A detailed investigation of this process is beyond the interests of this thesis, so refer to [Lodato, 2008] for a brief review of the topic. We'll however discuss how the magnetic field can transport angular momentum in a non-dissipative fluid intuitively: in such conditions, the field

CHAPTER 2. ACCRETION DISCS DYNAMICS

lines are bounded to the fluid elements, which they follow in their rotating motion. If we consider two fluid elements A and B linked by a field line and slightly separated in the radial direction, as illustrated in Fig. 2.3, they'll rotate with different velocities, causing a stretching of the field line thus generating a magnetic tension. This tension will slow down A and speed up B, transferring the angular momentum from A to B. Since the orbits are determined by the value of the angular momentum, A will move inwards, whilst B will reach an outer orbit, so that the distance between the elements will grow, generating a positive feedback mechanism resulting in instabilities.

Recent simulations have shown that the non-resistive fluid condition is very important to guarantee the effectiveness of this process. This leads to a problem when trying to apply it to YSO since, in such systems, the gas is generally not sufficiently ionised to be taken as non-dissipative (except in the innermost regions of the disc), so the observed amount of angular momentum transport cannot be justified and other sources of turbulence are needed.

2.2.3 Gravitational instabilities and self-regulation

Gravitational instabilities could be the solution to this issue: as already highlighted in sect. 2.1, self-gravity can even affect relatively light discs' dynamics, and it certainly plays a crucial role in the first stages of the system formation, when the mass balance is strongly in favour of the disc. In the next chapter, we'll examine in depth this and other effects of self-gravity in protostellar discs.

Let's now focus on the development of gravitational instabilities. From Eq. 2.12, it is evident that gravity is balanced by pressure gradients and the centrifugal force due to the rotational motion. Pressure is effective in stabilizing small wavelength perturbations, as it can be easily evinced from the direct dependency on k ; physically, this can be justified considering that over a small scale even minor variations of the pressure can originate important gradients, preventing the system from collapsing under the action of its own gravity.

On the other hand, rotation dominates in the opposite regime since, if the perturbation covers a large scale, the variances in velocity and angular momentum of different parts of the disc will become important and won't allow a free change of orbit necessary for matter to collapse, contrasting gravity.

From the simple solution of the inequality $(\omega - m\Omega)^2 > 0$, we thus find out such a system is stable if $Q \doteq \frac{c_s \kappa}{\pi G \Sigma} > 1$. Since κ is usually of the same order as Ω , we expect the disc to be in a condition of marginal stability when the disc's self-gravity starts affecting the vertical structure of the disc, as anticipated in the introduction of this section. From the analysis of the vertical component of the Navier-Stokes equations proposed in sect. 2.1, we have indeed derived the typical thickness of self and non-self-gravitating disc, whose ratio is exactly $\frac{H_{sg}}{H_{nsg}} = \frac{c_s \Omega}{\pi G \Sigma}$.

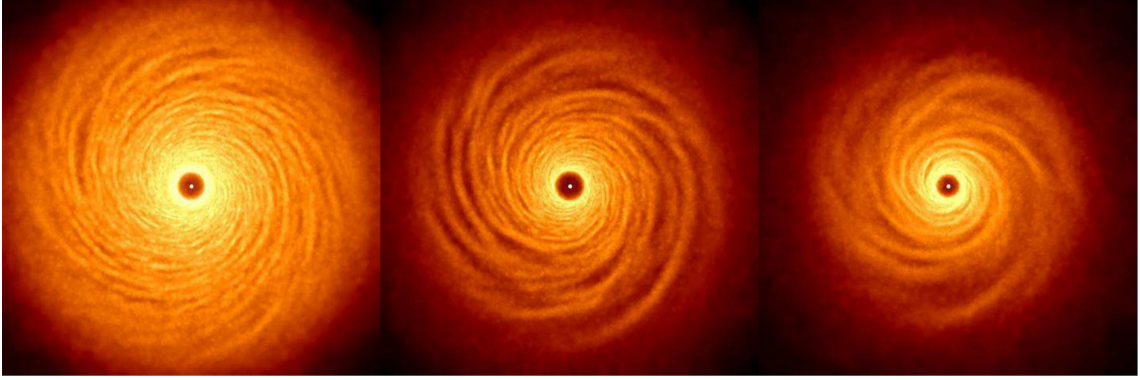


Figure 2.4: Surface density structure for discs with $M_D/M_* = 0.05, 0.1, 0.25$. Images from simulations performed by [Lodato, 2007].

The most unstable wavelength is the value that minimizes the right-hand side term, that is $k_0 \doteq \frac{\pi G \Sigma}{c_s^2} = \frac{1}{H_{sg}}$; from this relation, we can justify the application of the WKB approximation to thin discs, as $k_0 R = \frac{R}{H_{sg}} \gg 1$, hence $\tan \psi \ll 1$. It is important to note that, since the wavelength of the perturbations is linked to the disc thickness, more massive (thick) discs tend to be more unstable for large wavelength perturbations and the tight winding approximation holds only for small m , since k can be very tiny, resulting in more open spirals, as shown in Fig. 2.4.

Moreover, Q is proportional to c_s , which is proportional to the temperature of the gas; if no other sources of heat are present, an initially hot (stable) disc will cool down, ergo it will become unstable. At this point, instabilities will induce compression and shocks which dissipate energy, so the disc will heat up again and go back to a stable state. This way, the instabilities self-regulate, maintaining the disc in a condition of marginal stability characterized by a clear slowly-evolving spiral structure with finite amplitude and a Q -value fluctuating around 1.

Since in this condition the disc is only marginally unstable, we expect only the most unstable modes (with $k \approx k_0$) to be excited. Moreover, from Eq. 2.12 we grasp that for these modes, $Q \approx 1$ implies $\Omega_p \approx \Omega$, so all the excited modes are anticipated to lie near co-rotation.

For this mechanism to be efficacious, the cooling timescale needs to be not too small, since otherwise the heat produced by the instabilities, which evolve on the disc's dynamics timescale (Ω^{-1}), will be removed too rapidly and the disc will fragment (some areas will collapse). Numerical simulations reveal that proper values to prevent fragmentation are $\tau_{cool} \approx 3\text{-}13 \Omega^{-1}$, depending on the specific conditions of the simulation. In Chapter 3 more details about some of these simulations will be reported.

2.2.4 The α -prescription

Modern theories try to describe the transport of angular momentum due to turbulence in terms of an "effective viscosity" regulated by an unknown parameter α , which can be brought in to relate the vertically integrated pressure (Σc_s^2) and stress tensor (whatever its real nature is)

$$T_{R\varphi} = \frac{d \ln \Omega}{d \ln R} \alpha \Sigma c_s^2, \quad (2.17)$$

or equivalently ν and H in the case of classical viscosity ($\nu = \alpha c_s H$).

This approach was proposed by Shakura & Sunyaev in 1973 and was purely inspired by dimensional consistency: it is just a parametrization that simply moves the problem from knowing the origin of the kinematic viscosity to understanding the origin of α . In their work, Shakura & Sunyaev assumed the turbulence to be subsonic (otherwise shocks would originate, dissipating the perturbation) and isotropic, so that the maximum dimension of the eddies would be the thickness of the disc. Since the kinematic viscosity can be defined¹⁶ as $\nu = v \lambda$, where v is the velocity of the eddies and λ is their typical length scale, we deduce that $\alpha < 1$.

It is important to remark that this parameter is not necessarily a constant, but this constraint holds for every region in the disc. Less loose constraints and estimations can be inferred comparing observations to theoretical models, whose predictions are obtained using numerical simulations (we'll look at an example right then). For protostellar discs, a typical value is $\alpha \approx 0.01$. Let's moreover note that in a Keplerian disc $d \ln \Omega / d \ln R = -3/2$, so that the stress is negative; as described by Eq. 2.10, under these conditions it therefore opposes rotation, allowing gas to flow inwards and accrete while transporting angular momentum outwards.

Adopting this prescription and carrying out a linear analysis as described in the previous section, we can hence introduce a stress tensor describing the action of self-gravity: assuming the velocity and the self-gravitating potential (Φ) in the Euler equation to be made up of a quasi-steady component ($\mathbf{v}, \Phi_0$) and a rapidly fluctuating one due to a non-necessarily axisymmetric perturbation¹⁷ ($\mathbf{u}, \Phi'$), with analogous calculations as for Eq. 2.10, we can derive the angular momentum conservation equation:

$$\frac{\partial}{\partial t} (\Sigma R^2 \Omega) + \frac{1}{R} \frac{\partial}{\partial R} (\Sigma R^3 \Omega v_R) = -\frac{1}{R} \frac{\partial}{\partial R} \left[R^2 \Sigma (\langle u_R u_\varphi \rangle + \langle u_R^g e_\varphi^g \rangle) \right]. \quad (2.18)$$

¹⁶This definition comes in analogy to the definition of ν in the context of classical viscosity due to thermal molecular motion; this way, ν is understood as the effectiveness of viscous forces in transporting momentum.

¹⁷Note that here, the origin of the initial perturbation can be of any kind, seeing that the gravitational potential can be affected by any variation of the mass distribution. We're only investigating how self-gravity will act in response to this perturbation.

CHAPTER 2. ACCRETION DISCS DYNAMICS

Here the angular brackets indicate azimuthal and vertical averages and $\mathbf{u}^g \doteq \frac{-\nabla\Phi}{\sqrt{4\pi G\rho}}$. As in Eq. 2.10, the right-hand side represents a torque, which can be interpreted as due to an effective viscous force described by

$$T_{R\varphi} = -\Sigma(\langle u_R u_\varphi \rangle + \langle u_R^g e_\varphi^g \rangle) = T_{R\varphi}^{Reyn} + T_{R\varphi}^g \quad (2.19)$$

or, equivalently, by

$$\alpha_{sg} = \left| \frac{d \ln \Omega}{d \ln R} \right|^{-1} \frac{\langle u_R u_\varphi \rangle + \langle u_R^g e_\varphi^g \rangle}{c_s^2}. \quad (2.20)$$

As we will discuss in sect. 2.3, this effective viscosity can be associated with energy dissipation, thus to a heating rate for the disc. At thermal equilibrium, this rate, which can be expressed as a function of α thanks to the First Law of Thermodynamics¹⁸, must compensate for the cooling rate of the gas, so an expression for α in terms of this cooling timescale can be derived:

$$\alpha = \left| \frac{d \ln \Omega}{d \ln R} \right|^{-2} \frac{1}{\gamma(\gamma - 1)\Omega\tau_{cool}}, \quad (2.21)$$

where γ is the ratio between the specific heats and τ_{cool} is the cooling timescale¹⁹.

Once the angular velocity has been measured from observations of a real system and τ_{cool} has been estimated, a value of α can be computed and compared with that obtained from numerical simulations of disc dynamics using Eq. 2.20. The comparison shows reasonably good agreement.

Using Eq. 2.20, we can also relate the critic value $\tau_{cool} \approx 3\text{-}13 \Omega^{-1}$ to a maximum value for α_{sg} at which the disc will fragment, and interpret this threshold as an upper bound for the angular momentum that can be transported by self-gravity. In particular, this relation is derived under the approximation of a quasi-Keplerian disc²⁰ in a steady state.

Numerical simulations²¹ executed using different models and values for γ all return a threshold value of 0.06.

¹⁸ $d\varepsilon = -p d\nu = -p d\left(\frac{1}{\rho}\right) = \frac{p}{\rho^2} d\rho$. Integrating this relation with an appropriate expression for $p(\rho)$ and considering the rate of energy dissipation given by viscous-like forces, we can define a thermal timescale as a function of α .

¹⁹Careful considerations on the cooling processes regulating the observed system are made to estimate this timescale.

²⁰As mentioned at the beginning of sect. 2.1, this is the most common case for protostellar discs.

²¹Check [Lodato, 2007] sect. 6 for a brief report of the results, or [Rice et al., 2005] for the original paper.

CHAPTER 2. ACCRETION DISCS DYNAMICS

It's worth emphasizing the fact that this is a local analysis of the problem, because α is introduced to relate the stress tensor to the local pressure, ergo the angular momentum transport is attributed to the stress originating between neighbouring areas of the spiral structure. As we will notice in Chapter 3, simulations indicate that this approach is justified only if $H/R \approx M_D/M_* \lesssim 0.1$, thus this argument can't be applied to massive discs, which can in fact develop violent global instabilities transporting angular momentum and even energy much more rapidly. On the other hand, for light discs both local and global effects can emerge, but the latter are at least one order of magnitude lower.

2.3 Spectral energy distribution (SED)

Let's now consider classical viscosity again. Since it is a dissipative force, the gravitational potential energy released as the gas falls into the potential well of the central object and accretes is converted into heat²².

Such a dissipation can therefore be expressed in terms of the stress tensor, because the power dissipated in a rotational motion is given by the product of the torque and the angular velocity: dividing the disc into infinitesimal annuli of radial extension ΔR , the net torque exerted on an annulus at radius R by the neighbouring ones is simply $\frac{\partial G(R)}{\partial R} \Delta R$. Exploiting Leibniz's rule and considering that $\frac{\partial(G\Omega)}{\partial R}$ gives no contribution to the local heating of the disc as it is responsible for the transport of energy towards its outer edge²³, the power dissipated per unit area can be written as

$$D(R) = \frac{1}{2\pi R \Delta R} \frac{\partial \Omega}{\partial R} G \Delta R = R T_{R\varphi} \frac{\partial \Omega}{\partial R} = \nu \Sigma \left(R \frac{\partial \Omega}{\partial R} \right)^2. \quad (2.22)$$

At first approximation, this dissipation can be treated like blackbody radiation; from the Stefan-Boltzmann law²⁴, we can therefore derive the temperature of the disc, finding that it depends on the distance from the central object:

$$\tilde{T}_{\text{eff}}(R) = \left(\frac{1}{2\sigma_B} R \frac{\partial \Omega}{\partial R} T_{R\varphi} \right)^{1/4}. \quad (2.23)$$

As a consequence, the full SED is the superposition of different blackbodies, a "stretched blackbody" spectrum.

²²If this process dominates the disc's emission, we label the disc as *active*; otherwise, if the emission is dominated by the star's reprocessed luminosity (reflected or absorbed and re-emitted), we call it *passive*.

²³As recognised above in Eq. 2.10, a term like this is a divergence, thus it describes transport.

²⁴ D matches the definition of the flux recalled in Appendix B along with the properties of blackbody emission.

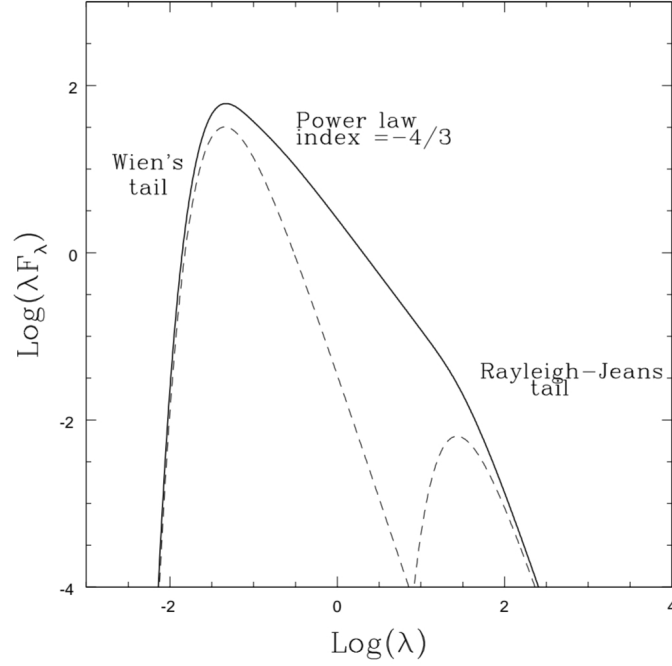


Figure 2.5: SED of a Keplerian, optically thick disc. The longer wavelengths correspond to colder, outer regions of the disc. Figure from [Lodato, 2007].

When considering an effective viscosity as in the previous section, however, we haven't proved that it really acts as a local dissipative force yet, since energy is rather transferred to the fluctuations than directly dissipated. We then have to make sure that the amount of energy transferred is identical to the one that would be dissipated by classical viscosity and verify that the fluctuations are then dissipated locally.

To address these problems we can adopt a Hamiltonian formulation²⁵ and relate the energy and angular momentum densities ($\mathcal{E}, \mathcal{L}$) transported by the waves to the wave action density $\mathcal{A} = \frac{\mathcal{E}}{\omega} = \frac{\mathcal{L}}{m}$, so that $\mathcal{E} = \Omega_p \mathcal{L}$. In the case of quasi-stationary waves ($\Omega_p = \text{const}$), we obtain that the transferred power per unit area is

$$\dot{\mathcal{E}} = D(R) = \Omega \dot{\mathcal{L}} + (\Omega_p - \Omega) \dot{\mathcal{L}}, \quad (2.24)$$

where $\dot{\mathcal{L}}$ is the most general expression of the torque, which can then be rewritten in terms of the stress tensor like in the previous case²⁶.

We therefore discover that, even in the simplest case of quasi-stationary perturbation, extra terms arise in the relation between the energy flux and the stress tensor,

²⁵Check [Cossins et al., 2009] sect. 2.3 for a more complete derivation.

²⁶Actually, a subtlety regarding the proportionality of the generalised stress tensor to the rate of strain $R \frac{\partial \Omega}{\partial R}$ has to be taken into account, if we want to proceed with an analogous derivation as above.

CHAPTER 2. ACCRETION DISCS DYNAMICS

which are no longer proportional through Ω except in the special case of co-rotation. If the perturbation is dissipated far from this condition, the second term describing global transport determines a great difference with respect to the dissipation purely due to viscosity. The amount of global transport can be in turn evaluated by the parameter $\xi = \left| \frac{\Omega_p - \Omega}{\Omega} \right|$.

This different behaviour can be converted into a distinction between light and massive discs suggested by numerical simulations: for the former, the extra terms are usually much lower than the classic viscous one, since the majority of the modes excited by the cooling process generally interact with one another and dissipate over short length scales; however, some modes can last longer (especially the ones with higher wavelengths), propagating to further distances and generating the observed deviation from a pure local behaviour. Whereas, for massive discs, global effects result in perturbations showing huge variations of α over time, up to 0.5, causing a fast redistribution of the gas within the disc and thus a "global" angular momentum and energy transport.

Moreover, in some cases, the simple argument exposed at the beginning of this section does not hold because up to now we've ignored several variables:

- I. A more detailed dissection including the optical properties of gas and dust should be performed.
- II. We've already described in sect. 2.2.3 the outcomes of the self-regulating mechanism in unstable self-gravitating discs, where the rotation curve becomes flatter and temperatures are higher at large radii since there the disc self-gravity successfully overcomes the central object's one. As a result, the SED of these discs is expected to be flatter and to display a secondary bump at large wavelengths.
- III. The heating due to the radiation coming from the star is often non-negligible; generally, both the "active" and the "passive" heating are present in a disc, and their relative importance is regulated by the comparison between the star accretion rate and the stellar luminosity.

Despite all this, temperatures associated with discs are much lower than a star's photosphere one, so the full spectrum of a star-disc system will manifest an excess at near and mid-infrared wavelengths when compared to a single star SED; this characteristic is exploited both to spot and to classify these systems, as we will envision in the next chapter.

Chapter 3

Protostellar discs

3.1 Confirmation of theoretical expectations

Let's begin this chapter by reporting the results of some numerical simulations run by [Cossins et al., 2009] that have almost completely validated the theoretical previsions presented in Chapter 2 about gravitationally unstable protostellar discs.

All the images in this section come from the article cited above.

We are dealing with 3D *smoothed-particle hydrodynamics* (SPH, [Price, 2012]) simulations; in such simulations, the disc is made up of a fixed amount of identical particles which do not represent the real molecules present in the disc but allow a description of the global fluid-like behaviour. The code follows the evolution of each particle, and the global physical quantities of the disc are computed as sums or averages over the individual particle values.

The setup details can be found in [Cossins et al., 2009], sect. 4; let's only report what's meaningful for our purposes:

1. The central object was modelled as a point mass onto which the particles accrete if they get close enough and match other specific conditions.
2. The disc was taken to be gaseous and non-magnetized.
3. The gas was modelled as a perfect gas with $\gamma = 5/3$.
4. The real nature of the cooling process in accretion discs is not known, but it can be simply modelled by the parameter τ_{cool} , absorbing in its evaluation all the physical details. In these simulations, the product $\Omega\tau_{cool} = \beta$ was assumed to be a constant within the whole disc, and different simulations were run varying this value.
5. For a fixed value of β , various simulations were performed varying the ratio $M_D/M_* = q$.
6. The simulations were three-dimensional, so finite-thickness effects arose.
7. The simulations lasted enough long so that the disc settled in a quasi-steady

CHAPTER 3. PROTOSTELLAR DISCS

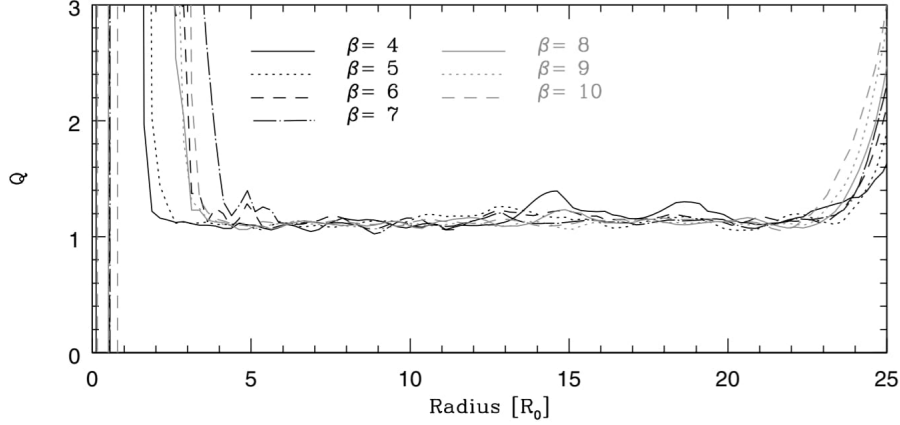


Figure 3.1: Profiles of Q against radius for different values of the cooling timescale β and $q = 0.1$. The radius is given in specific code units.

marginally stable state.

The first important confirmation coming from these simulations is that the self-regulating mechanism is effective and stabilizes the disc to a value of Q close to unity over a large range of radii (view Fig. 3.1) if $\beta \geq 5$; otherwise, the disc will fragment, as predicted in sect. 2.3.

The stabilization is also monitored by studying the amplitude of the perturbations $\frac{\delta\Sigma}{\Sigma}$: the simulations correctly attest that it saturates to a value which increases with q and decreases with β , creating the quasi-steady spiral structures shown in Fig. 3.2, where the effects of the variation of the parameters are very clear. Since the rotation of the discs in these pictures is anticlockwise, the simulations further corroborate that the excited modes are trailing waves, as anticipated.

Furthermore, in the context of WKB approximation, an expression for the action $\mathcal{A}$ introduced in sect. 2.2.3, and therefore of the energy density, in terms of $\frac{\delta\Sigma}{\Sigma}$ can be found. Comparing it to the cooling process one, since they must be equal at thermal equilibrium like done in sect. 2.2.4, we obtain:

$$\frac{\delta\Sigma}{\Sigma} = \left(\frac{2}{\epsilon\beta\gamma(\gamma-1)\mathcal{M}\tilde{\mathcal{M}}} \right)^{1/2}, \quad (3.1)$$

where ϵ is an ideality factor accounting for the amount of energy effectively converted into heat, $\mathcal{M} = \left| \frac{m\Omega_p}{kc_s} \right| = \frac{v_p}{c_s}$ is the wave phase Mach number, and $\tilde{\mathcal{M}}$ is the Doppler-shifted phase Mach number.

Such a dependency on β is confirmed by the simulations.

Carrying out a Fourier analysis of the wave modes, it was found that azimuthal ones' excitation is almost independent from β and the radius, while the disc to the

CHAPTER 3. PROTOSTELLAR DISCS

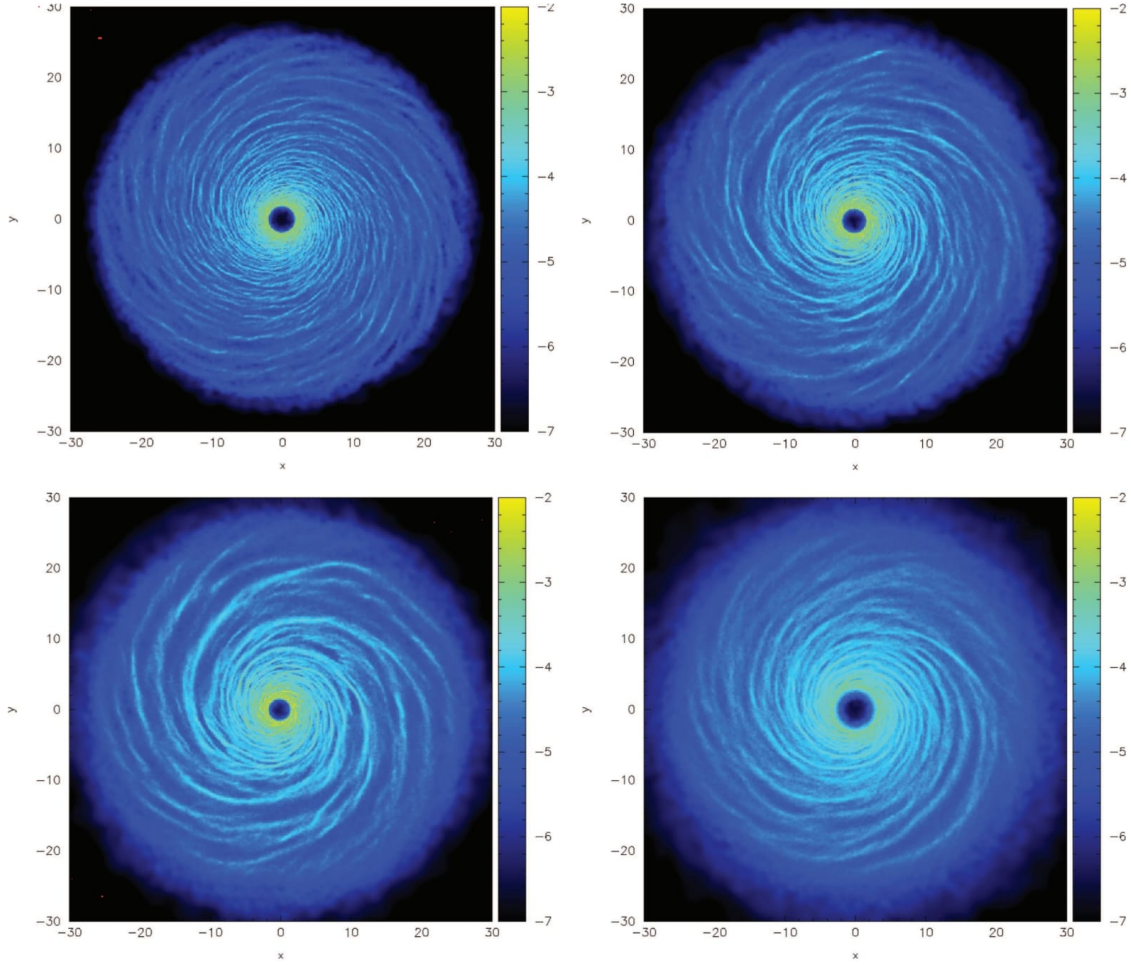


Figure 3.2: Surface Density structures for discs with $\beta = 5$ and $q = 0.05$ (top left panel), 0.125 (top right) or 0.1 (bottom left), and with $\beta = 10$ and $q = 0.1$ (bottom right). The logarithmic scale is in specific code units.

The disc rotates anticlockwise so all the excited modes are trailing waves and it's also evident that more massive discs develop structures with fewer and more well-defined arms, while the only effect of the cooling timescale is a decrease in the amplitude.

central object mass ratio has a significant impact: as anticipated in sect. 2.2.3 and displayed in Fig. 3.3, more massive discs tend to excite modes with lower m .

On the other hand, although radial modes maintain the same dependency on β and q as expected, they exhibit a correlation with the radius; in particular, the peak wavelength increases with the radius, and this is consistent with the prediction of the most unstable wavelength $\lambda_0 = H/2\pi$ (cf. sect. 2.2.3), as the thickness of the disc increases with the radius ($H(R) \approx RM_D(R)/M_*$ and the total mass of the disc within a

CHAPTER 3. PROTOSTELLAR DISCS

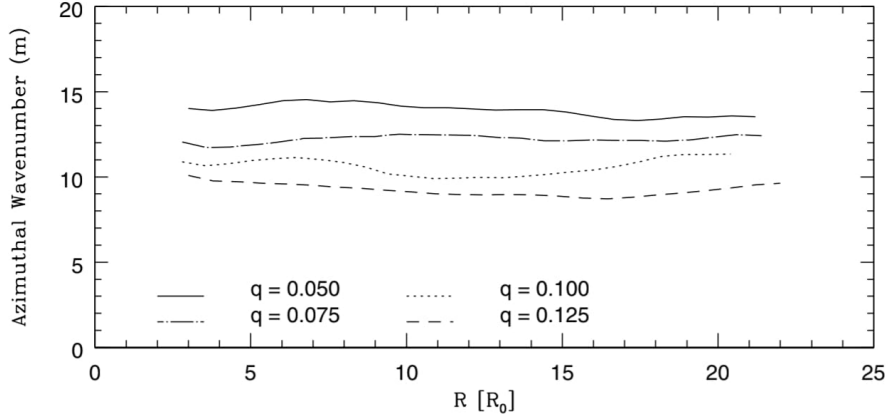


Figure 3.3: Variation of the average azimuthal wavenumber excited as a function of the radius for different values of q and $\beta = 5$.

given radius obviously increases with R). The plot in Fig. 3.4 clearly evidences that the above relation holds for all R . These results, moreover, emphasize that the radial spacing of the arms only depends on the surface density and on the temperature, since $k \approx H_{sg}^{-1} = \frac{\pi G \Sigma}{c_s^2(T)}$.

These simulations verified that the excited modes are always close to co-rotation at every radial distance, too. This was done by calculating $|\Omega_p - \Omega|$ from the dispersion relation (2.12)¹ using the average m and k found from the Fourier analysis.

Besides, using these data, the anticipation about the value of the Doppler-shifted phase Mach number made at the end of sect. 2.2.3 has been borne out. The fact that $\tilde{\mathcal{M}} \gtrsim 1$ in all the simulations therefore confirms that in all these cases the perturbations are tightly wound, and this is also supported by the plot in Fig. 3.5, from which the prediction that the openness of the spiral structure increases with the mass of the disc can be verified too, attesting in addition that the tight winding approximation breaks down if $M_D/M_* \gtrsim 0.1$, because over such a threshold ψ is no longer small.

It was moreover possible to precisely quantify the amount of global transport through the parameter ξ , since it can be simply rewritten as

$$\xi = \frac{|\Omega_p - \Omega| m}{k c_s} \frac{k c_s}{m \Omega} = \tilde{\mathcal{M}} \frac{k H}{m} \approx \frac{1}{m \cos \psi}, \quad (3.2)$$

where the last relation holds because the average radial wavenumber k is very close to the most unstable value, as seen above. Such a relation justifies the link between non-locality and the openness of the spirals and, consequently, the mass of the disc. The

¹Actually, as said at the beginning of this section, since the simulations are three-dimensional, the equations derived in Chapter 2 need to be corrected. In particular, solving the Poisson equation for the specific configuration of the disc density adopted in these simulations, we find that the self-gravity term in the dispersion equation must be divided by $1 + |k|H$.

CHAPTER 3. PROTOSTELLAR DISCS

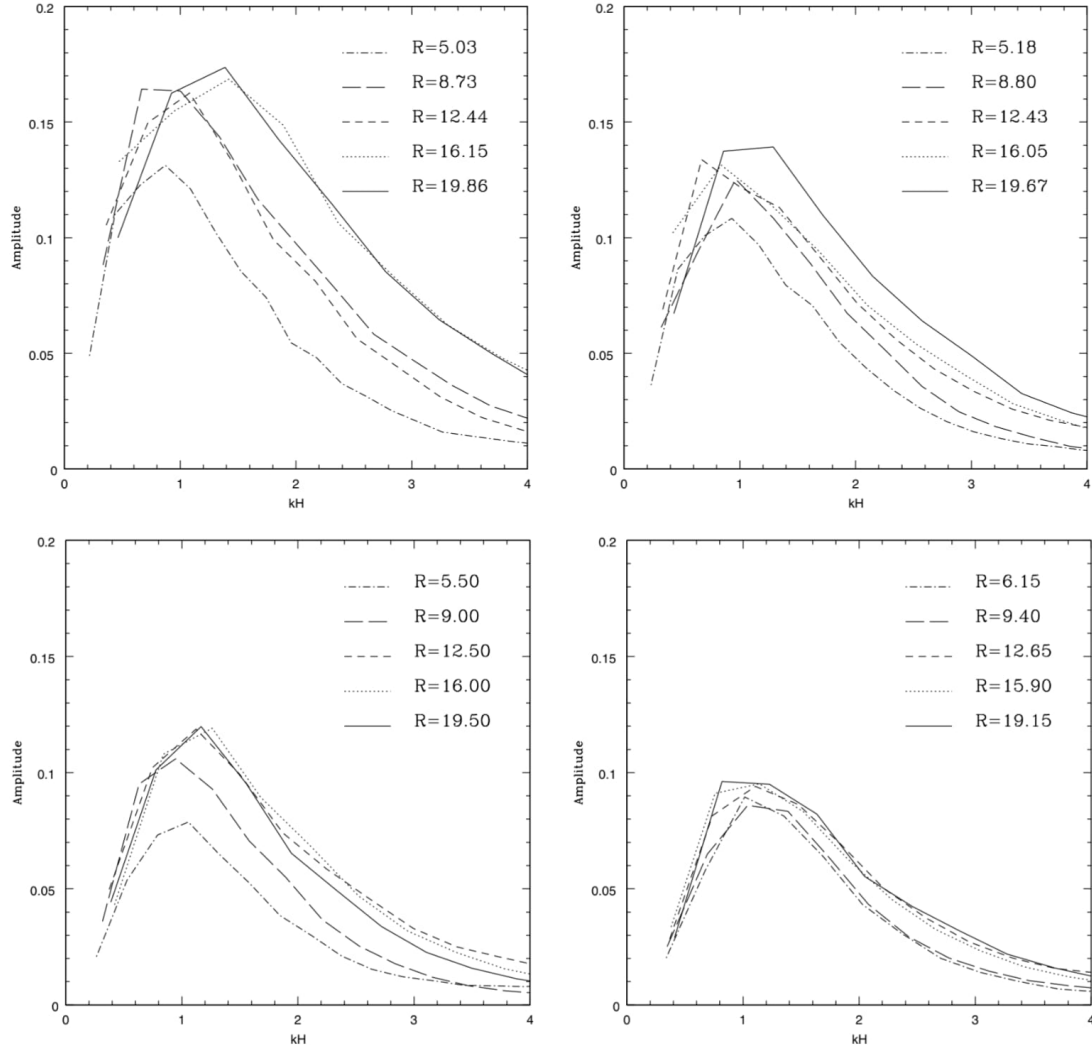


Figure 3.4: Modes amplitude plotted against the normalized wavenumber kH at different radius for $\beta = 4$ (top left panel), 6 (top right panel), 8 (bottom left panel) and 10 (bottom right panel).

As expected, the peak at $kH = 1$ is visible in all the cases.

simulations confirmed that the importance of non-local effects increases with mass and that for low-mass discs, although such a contribution is not exactly negligible (being about one order of magnitude lower than the local one, as reported in Fig. 3.6), the transport of angular momentum can be well parametrized by a local α -prescription, because the waves are always close to co-rotation.

Finally let's notice that the ideality factor is found to be around 20% in all the simulations, meaning that not all the energy of the modes is effectively converted into

CHAPTER 3. PROTOSTELLAR DISCS

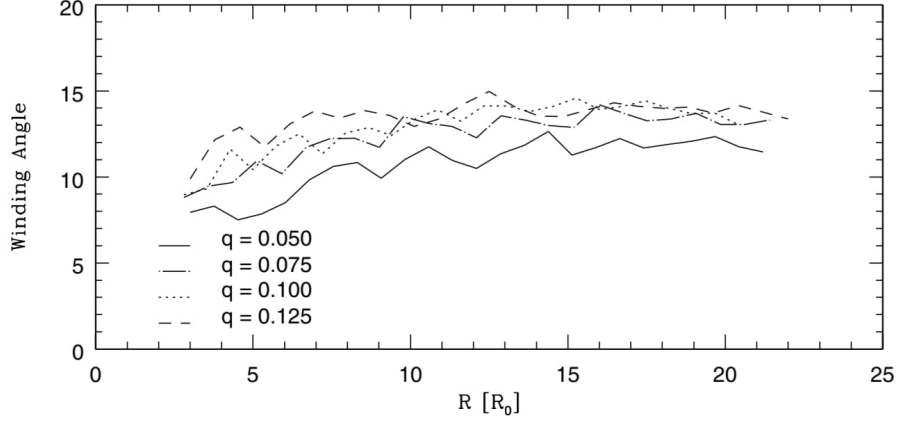


Figure 3.5: Pitch angle as function of radius for $\beta = 5$ and different values of q .

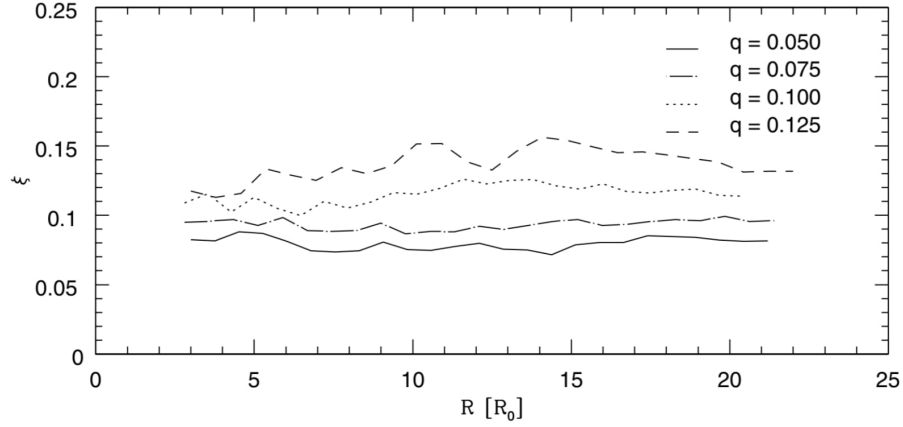


Figure 3.6: Non-local transport fraction as a function of radius for $\beta = 5$ and different values of q

heat. The matter is important for the consideration about the SED of these systems made back in sect. 2.3.

3.2 Formation, classification and observations

Stars develop in the core of turbulent molecular clouds; this core usually carries a specific angular momentum $j \sim 10^{21} - 10^{22} \frac{\text{cm}^2}{\text{s}}$, which can be estimated from its *jeans length*, the linear scale of a system in which gravity is balanced by pressure², and the measurements of the angular velocity of the gas in the core. This value is several

²Consult [[Lodato, 2008](#)] for the explicit derivation.

CHAPTER 3. PROTOSTELLAR DISCS

orders of magnitude larger than the typical value for a star³.

To form a star⁴, the gas is therefore forced to lose angular momentum, and a disc-shaped object is the natural outcome of this process: as a rotating gas cloud collapses under its own gravity, the contraction is not uniform because the centrifugal force due to rotation varies with position. It is strongest in the plane perpendicular to the rotation axis that roughly divides the cloud into two equal parts (the midplane) so, in this region, collapse is halted when the gas reaches a precise limit orbit (which depends on the shape of the gravitational potential). Away from the midplane, however, the centrifugal force gradually fades, so collapse is allowed up to smaller radii. In particular, material continues to fall inward along the polar directions, where the centrifugal force vanishes, ultimately feeding the formation of a protostar at the centre. Once the protostar forms, the gravitational potential of the system changes, slightly altering the orbit of the gas in the midplane, which now settles to a nearly circular Keplerian orbit at a radius $R_0 = \frac{j^2}{GM}$ (where M is the mass of the already formed protostar), forming a ring. For a star like our Sun, $R_0 \approx 10^2$ - 10^4 AU.

The evolution of this gas ring is fully determined by Eq. 2.11, and it can be proved that it consists of a simultaneous inward and outward spreading (at least in the case of constant ν - check [Lodato, 2007], sect. 2.6), resulting in the formation of a disc. This disc is, at first approximation, still Keplerian, so the angular velocity decreases as a function of the radius, leading to the desired rearrangement of angular momentum, as it flows outwards according to Eq. 2.10, allowing at the same time the gas to move inwards and feed the central protostar (main infall phase). In these first stages, the disc is still wrapped in a gas cloud that continuously collapses onto the initial ring, accreting the disc mass and partially compensating for the losses due to the infall process.

The infall is generally more rapid than the disc's accretion, so the importance of the disc self-gravity will gradually fade away, and the system will switch to a state dominated by the star's gravity. As explained in the previous chapter, such systems show strongly mass-dependent properties, and are hence grouped on the basis of the stages of the process described above. In particular, the fundamental parameter to differentiate the classes described below is their SED, which exhibits an excess at longer wavelengths (with respect to a standard stellar photosphere) whose magnitude decreases as the system evolves, both because the gas envelope progressively disappears and the effective emission of the disc and the star becomes visible, and because the system becomes less gravitationally unstable.

³Our sun has a specific angular momentum of $10^{16} \frac{cm^2}{s}$.

⁴We are ignoring the effects of interactions between forming stars that could lead to a binary system and also magnetic field effects, but the process described is valid, anyway.

3.2.1 Class 0 and I systems

These are the youngest objects, still enveloped by the gas of the original cloud progressively accreting the disc, which is, in turn, extremely massive. These systems have been associated with the observations of very massive stars ($M_* > 8 M_\odot$), which enter the principal sequence while still accreting due to their large mass.

The presence of the gas envelope makes direct observations difficult, so there remains substantial uncertainty about the presence and the characteristics of the disc in the most massive and wide systems (O-type protostars, with masses over $20 M_\odot$). Smaller systems (up to an extension of 10^3 AU), where the central object is a B-type star ($8-20 M_\odot$), are instead recognised as proper discs and some more precise measurements can be performed. For example, an important parameter is the outflow rate from the inner parts of the disc (the star's accretion rate); its value is of the order of $10^{-4}-10^{-3} M_\odot/\text{yr}$, and it can be theoretically justified considering global angular momentum transport due to frequent brief spiral structures that originate in massive discs and are related to large α values, as described at the end of sect. 2.2.4.

Abrupt increases in the accretion rate are identified as the cause of great outbursts, with a corresponding enhancement in luminosity by a few orders of magnitude, observed in a group of stars labelled *FU Orionis objects*. These objects are understood to be quiescent Class I systems showing transient intense activity, and data from the observations are compatible with the hypothesis of self-gravitating discs with a mass of the same order of magnitude as the central object's one. The variation of the accretion rate could hence be imputed to the development of gravitational instabilities, as suggested by the results of numerical simulations briefly commented on at the end of the previous chapter, but a comparison with data from observations still lacks.

The more accepted explanation is, in fact, given in terms of thermal-viscous (*limit cycle*) instabilities: in the general case of ν dependent on Σ , Eq. 2.11 becomes non-linear but some properties can still be deduced by rewriting it in terms of $\mu = \nu\Sigma$ which is essentially the mass flowing out of a disc annulus at radius R (since all the quantities are functions of the radius). If $\frac{\partial \mu}{\partial \Sigma} < 0$, indeed, the equation is no longer a simple diffusion equation, as a decrease in the surface density is followed by an increase of the flux, resulting in a positive feedback mechanism for which more mass will flow away from the annulus. Solving the equation for the whole disc attests that when an instability is triggered at a certain radius as described above, a fast perturbation front will propagate inwards, ionizing the gas thus increasing the optical and infrared energy flux emitted by the disc, while a slow wavefront will move outwards until it reaches a zone where the surface density can be stable. The gas in the inner parts of the disc will be, therefore, consumed quickly (causing the observed outbursts), and will be slowly replaced by the outer materials, so the whole process will repeat after a long quiescent phase.

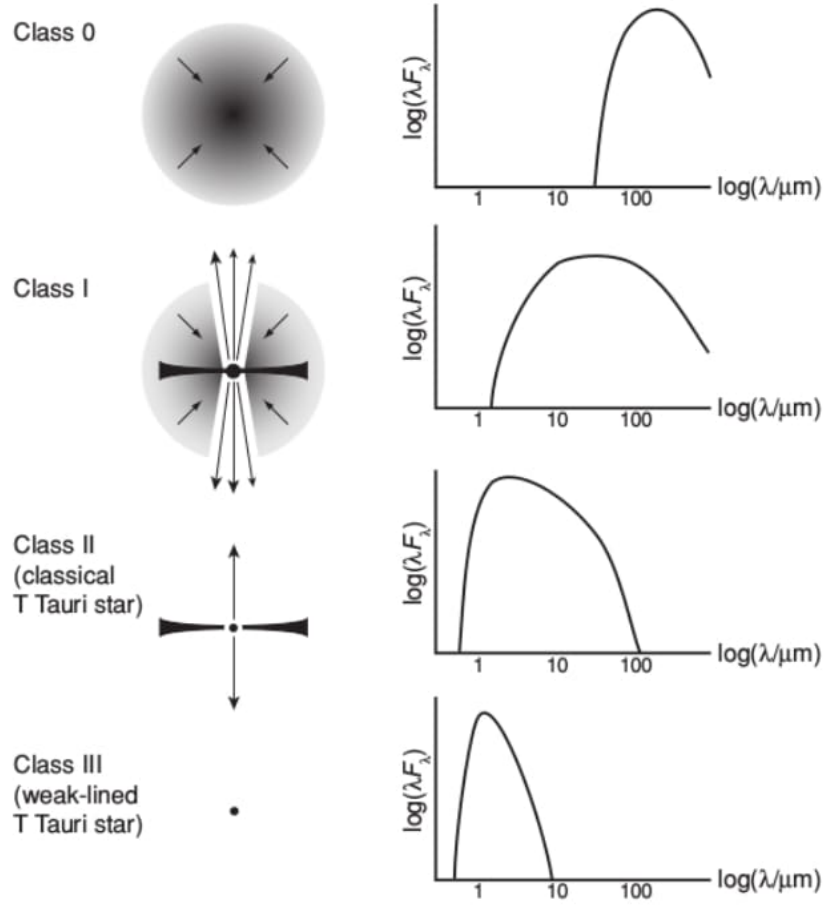


Figure 3.7: YSO classification. On the left, a schematic general representation of the object; on the right, the typical SED.

This mechanism is, however, not able to fully accomplish the observations (See [Lodato, 2008] sect. 7.1 for a detailed summary of the problems); this is why alternative explanations such as gravitational instability have been proposed.

3.2.2 Class II and III systems

Class II objects are also referred to as *Classical T-Tauri stars*. In these systems, the main infall phase has ended and the disc is visible. They are expected to be less extended than the previous ones and are generally identified with solar mass young stars. The observed systems attributed to this class can be both gravitationally stable and unstable, depending on the relation between the ratios M_D/M_* and H/R , where $H(R)$ can be estimated from the temperature of the disc, M_* from the star luminosity,

and $M_D(R)$ is obtained rescaling the mass derived from dust sub-millimetre emission, in the assumption of an optically thin medium and provided the opacity of the dust is known.

The last measurement, especially, is subject to systematic errors and caveats regarding the applicability of the approximations employed, since the opacity and the dust size distribution can be very different from the interstellar medium ones, due to the presence of large dust grains⁵; ergo, it is likely for the disc mass to be underestimated and consequently the system has more chances to be gravitationally unstable.

For these objects, the accretion rate onto the star can be determined from the emission lines of the gas which falls into the star following the stellar magnetic field lines (*magneto-sphere attraction*), or from variations of intensity in the photospheric spectrum attributed to accretion shocks on the star's surface.

Class III objects are the most evolved ones: the gas has almost completely vanished and what remains is a rocky *debris disc*. This stage is typically reached after a few million years, a lifetime which is estimated from the fraction of stars which display an excess in infrared emission in young stellar clusters of different ages.

3.3 Planet formation

Protostellar discs are of particular interest because they are the sites where planets assemble. More than for specific significances of the discs themselves, indeed, they are often studied in order to better understand this process and the origin of our own Solar System.

Up to now, thousands of extrasolar planets have been found, with large variability in terms of mass, eccentricity, and other parameters; most of the planets are found very close to the star they orbit around (less than 1 AU) but, actually, this is in large part a consequence of the intrinsic limitations of the detection methods used: the majority of them have been detected through the *transit* or the *radial velocity* methods⁶, which require being able to observe several orbits of the planet around the star to pick out a periodic pattern and extract the planet's features. As a result, only the closest planets, which have short orbital periods, are typically discovered. It's relevant to observe that many of these planets are too massive for having formed at such a distance, since the locally available mass given the disc's density is too small, suggesting that radial large-scale migrations can occur.

Trying to describe planet formation, two models have been developed; in both of them, the disc's self-gravity may play a crucial role.

⁵In sections 3.3.1 and 3.4 we will go deeper into some of the main properties of dust to justify this.

⁶Other classical methods used to spot exoplanets are *microlensing*, *direct imaging*, and *astrometry*. Examine [Kaushik et al., 2024] and [Currie et al., 2023] for some details about each method.

3.3.1 The core accretion model

This model is able to explain both the formation of Earth-like rocky planets and gas giants (data from Jupiter and Saturn observations suggest that these planets have a solid core, too) in terms of the gradual gathering of solid elements in the disc: dust grains first grow through inelastic collisions from micrometer to meter scales; at this point, a not yet fully understood mechanism leads to the growth of kilometre-sized bodies known as *planetesimals*; eventually, gravitational focusing among planetesimals drives runaway core growth. If the core is very massive, it can, moreover, attract a vast amount of gas, becoming a gas giant.

The intermediate phase in this process can't be explained neither in terms of inelastic scattering, since the grains are too big for it to be effective, nor by gravitational focusing, since they are too small to generate a significant gravitational potential. Ergo, a different mechanism needs to be found.

In this context, gravitational instabilities due to the disc's self-gravity become particularly intriguing, for spiral structures determine the arising of local maxima in the pressure (seen as a function of the radius). As a consequence of hydrodynamics equations presented in the previous chapter, pressure gradients modify the rotation curve of the gas, but do not affect the solid component's motion, which will maintain a Keplerian orbit and will therefore rotate more or less rapidly than the gas, depending on the sign of the gradient. Between the gas and the grains, a drag force will in turn develop, generating a radial motion of the solid chunks and concentrating them to the pressure maximum. For intermediate size pebbles, in particular, the movement induced by this force occurs on a very short timescale and is very effective, since the solid component is neither too light (in this case it would be strongly coupled to the gas and would simply follow its motion without any difference in velocity), nor too heavy (the radial velocity would be negligible). This is clearly shown in Fig. 3.8

This mechanism could, in principle, solve the problems exposed above, and numerical simulations are being built to verify it. For instance, consult [[Longarini et al., 2023](#)].

3.3.2 The gravitational instability model

As discussed in sect. 2.2.3, if the cooling rate of the disc is too high, the self-regulation mechanism is not efficient, and the disc will rapidly fragment into gravitationally bound clumps, leading to the formation of gas giant planets without a solid core.

Considering that the most unstable wavelength attributed to the instabilities is $\lambda = 2\pi H$ (cf. sect. 2.2.3), the mass of the fragment can be roughly estimated as $M_{frag} = \Sigma \lambda^2 = \frac{4\pi}{Q} \left(\frac{H}{R}\right)^3 M_*$. For example, for a typical unstable disc, with $Q \approx 1$ and $H/R \approx 0.1$, around a star like our sun, $M_{frag} \approx 12 M_{Jupiter}$.

It is therefore evident that this model is incapable of accounting for the existence of

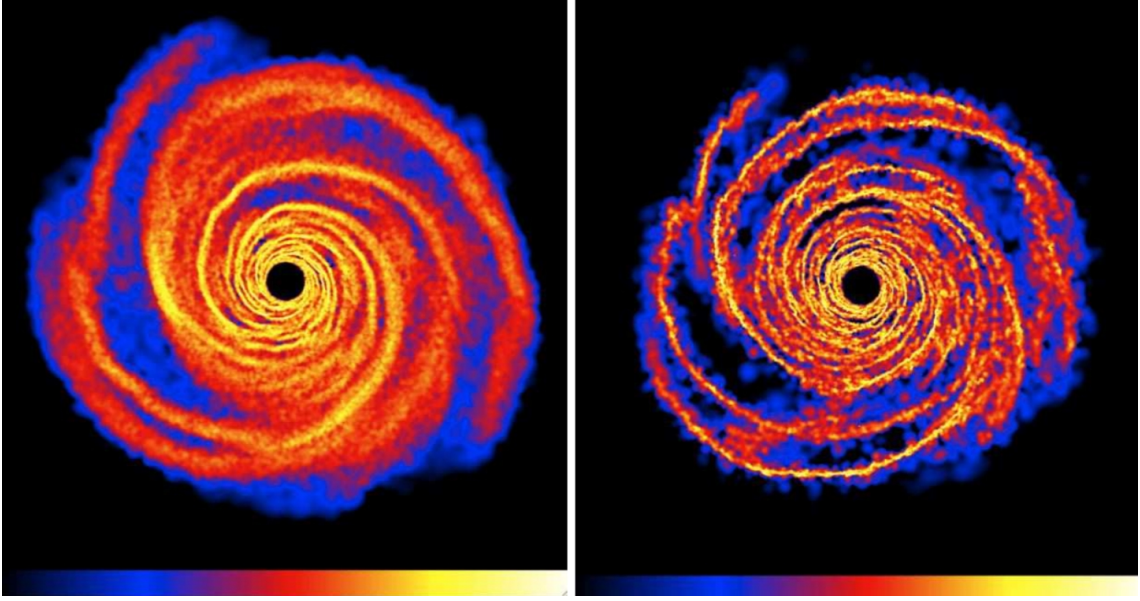


Figure 3.8: Surface density of the solid components in a self-gravitating gaseous disc. In the left panel the size of the components is 10 m , and the distribution is very similar to the gas one (compare with Fig. 2.4 for example). In the right panel the size is 50 cm and the drag effectiveness is evident.

Figure from [[Lodato, 2007](#)].

rocky planets, and can only explain the formation of the most massive planets or brown dwarfs. Moreover, the cooling rate is generally large enough to induce fragmentation only at considerable distances from the star (50 - 100 AU). Confirmations to this model are thus still lacking since, as said in the previous section, it is quite difficult to spot planets very far from their host star. Moreover, the real nature of the cooling process in accretion discs is not known yet, so the previous estimations are subject to considerable uncertainty.

3.3.3 Planets and instabilities

In the light of what is exposed above, the possibility of identifying gravitationally unstable discs from direct observation could be of great help in confirming the role of GI in planet formation and validating both of these models.

Moreover, the presence of a planet in a disc can give rise to another phenomenon which observations might enable us to characterize: the gravitational interaction between a disc and an embedded planet can indeed be an alternative source of instabilities leading to the formation of spirals.

CHAPTER 3. PROTOSTELLAR DISCS

This topic has already been studied extensively⁷, and it is well established that such spirals show quite different features with respect to GI-driven ones, evolving, for instance, with a constant pattern speed equal to the planet’s orbital velocity.

This trait is in significant contrast with the property of GI to be co-rotating, so the two kinds of instabilities are very likely to be distinguishable from direct measurements of the pattern speed.

3.4 Dust properties and dynamics

Let’s conclude this chapter by briefly presenting the properties of dust relevant for our purposes. Way more detailed and quantitative analyses on this topic can be found in [Longarini et al., 2023], [Kataoka et al., 2014], [Sierra and Lizano, 2020] and [Armitage, 2020].

Depending on the ordering between their dimension and mean free path, dust grains dynamics can be studied in two different regimes: in the *Epstein regime*, the size is small enough so that grains can interact with single gas molecules, while in the *Stokes regime* they are treated as particles immersed in a proper continuous fluid.

In both regimes, an expression for the drag force between the gas and the dust in terms of the difference of their velocities can be derived, and also a characteristic stopping time τ_s can be defined. The coupling between dust and gas can therefore be quantified by comparing such a timescale with the typical dynamical one, and this can be accomplished by introducing the *Stokes number*

$$\text{St} = \frac{\tau_s}{\tau_{dyn}} = \tau_s \Omega_k = K \frac{\rho_d a}{\rho_g H_g}, \quad (3.3)$$

where K is a constant that varies between the different regimes, ρ_d is the density of the dust, a is the size of the dust grain, ρ_g is the gas density, and H_g is the thickness of the gas disc.

The coupling is strong when $\text{St} \ll 1$, because in this case the dust rapidly modifies its motion under the influence of the drag force and follows the gas. Since the Stokes number increases with the grain size, the intuitive prediction that small pebbles are better coupled to the gas is justified.

Furthermore, it can be demonstrated that the solution of the equation of motion in the vertical direction for these grains is of the kind of a damped harmonic oscillator; hence, granules will eventually deposit on the midplane of the disc, where they will be even more coupled to the gas because the timescale introduced above is proportional to ρ_{gas}^{-1} and the density usually decreases when moving away from the midplane of the disc (cf. App. A.3.1).

⁷Check [Paardekooper et al., 2023]

CHAPTER 3. PROTOSTELLAR DISCS

The opacity of a system is determined by how its constituents absorb and re-emit radiation. In particular, it can be defined as $\kappa_\nu \doteq \frac{\sigma_\nu}{m}$ (cf. App. B), where σ_ν is the absorption cross-section and m is the mass of the particle considered. For a system made up of both gas and dust, we can therefore compute the two contributions using Mie theory for the scattering of electromagnetic waves onto spherical objects. For instance, in the sub-mm region of the electromagnetic spectrum, opacity is dominated by dust grains with typical dimensions varying between some micrometres and a millimetre.

Informations about dimensions and chemical composition of dust grains can be obtained from the study of spectral resonances between 3 and 100 μm and comparing the position of the peaks with the SED of the interstellar medium. This characterization is essential to estimate the opacity of the system, since great differences can be spotted in the two cases.

Chapter 4

Measuring pattern speed

In Chapter 2 we deduced that gravitational instabilities in protostellar discs are expected to be launched at co-rotation, a prediction later verified by numerical simulations. What remains unclear, however, is how the pattern speed evolves over time.

The simplest scenario is obviously that of constant velocity; if this was the case, inner regions of the spiral would rotate faster than the outer ones, since the Keplerian angular velocity scales as $R^{-3/2}$. The spiral would therefore wind up and eventually disappear in only a few orbital periods. Such an outcome is inconsistent with numerical simulations, since the spiral structures are found to last over much longer timescales.

This discrepancy arises because spiral arms are not static: as discussed in sect. 2.2.1 and 2.2.3, they are continually generated and dissipated through shocks. As a result, no simple analytical expression can describe a global, time-dependent pattern speed, as all the calculations proposed in the previous chapters are carried out in the linear regime in perturbation theory, while shocks are intrinsically non-linear effects.

Nevertheless, considering that the dynamical evolution of such systems far exceeds human timescales, it may still be possible to measure the instantaneous pattern speed directly from high-resolution observations of discs. The aim of this thesis is therefore to investigate whether ALMA images of protostellar discs contain sufficient information to constrain the velocity.

The present is an important step because the ability to measure pattern speed allows us to characterize the type of spiral observed, discriminating between GI or planetary spirals, for instance.

4.1 The hydrodynamical simulations and the MCFOST code

This study made use of data obtained from a suite of 3D SPH simulations of gas discs¹ previously executed by Dr. Longarini using the code PHANTOM [Price et al., 2018]. More precisely, these data consisted of snapshots of the column density distribution at different times.

In all the simulations, the cooling processes were described by the previously introduced parameter $\beta = \Omega\tau_{cool}$, fixed to a value of 10; the radius of the system was $R = 100AU$, and the mass of the central object was 1 solar mass. Different simulations were run varying the disc to star mass ratio $q = 0.1, 0.15, 0.25$, and 0.3 . In each simulation, 2 snapshots were recorded, one at the starting point and one at the tenth year of simulation. Such a time span was chosen because ALMA has been operating for about 10 years now, so the idea was to try to determine whether the pattern speed can be measured from the available real records. For some simulations, an extra snapshot at the fifth year was taken to try to perform more precise calculations.

Such data were employed as inputs for the radiative transfer code MCFOST [Perrin et al., 2018]. Along with the disc image, a parameter file had to be prepared to run the program. Table 4.1 reports the meaningful parameters that have been used.

Among the variety of outputs produced by the given code, our interest was towards the image of the disc, i.e. the map of the flux, which simulates how it would be observed in the sub-mm region of the EM spectrum. This image is produced via a ray-tracing method: the output intensities are obtained by integrating formally the source function² of the system, which is computed by averaging the information coming from a Monte Carlo simulation of photons' propagation. The high number of photons was chosen to produce high-resolution images.

MCFOST simulates the radiation emitted by a central star and how the disc reprocesses it; the final emission is thus that of a passive disc. More precisely, it simulates dust emission in the midplane. For ALMA observations on the systems considered, this approach is justified because, as pointed out at the end of the previous chapter, the opacity in the sub-mm is dominated by small-sized dust grains, which naturally tend to deposit in the midplane and have the strongest coupling to the gas. A rough

¹These simulations were very similar to the ones mentioned in the previous chapter; the numerical setup is also similar to the simulations used in [Longarini et al., 2023] with the important difference that dust was not included here.

²Cf. Appendix B.

CHAPTER 4. MEASURING PATTERN SPEED

Parameter	Description	Value
nbr_photons_eq_th	Number of photon packets launched when computing the temperature structure	1.28e7
nbr_photons_image	Number of photon packets launched when computing a disk image	1.28e8
grid (nx,ny)	Number of pixels for the output images	(512, 512)
size [AU]	Physical size of the produced maps	320
(RT_imin, RT_imax)	Minimum and maximum inclination to compute the Ray Tracing image	(0,0)
disc PA	Position angle of the disk major axis from North to East	0

Table 4.1: Parameters for the MCFOST code. All the other parameters could be ignored since the physics of the disc was already known (it's given by the hydrodynamics simulations), thus the other informations produced by MCFOST were not relevant for our purpose.

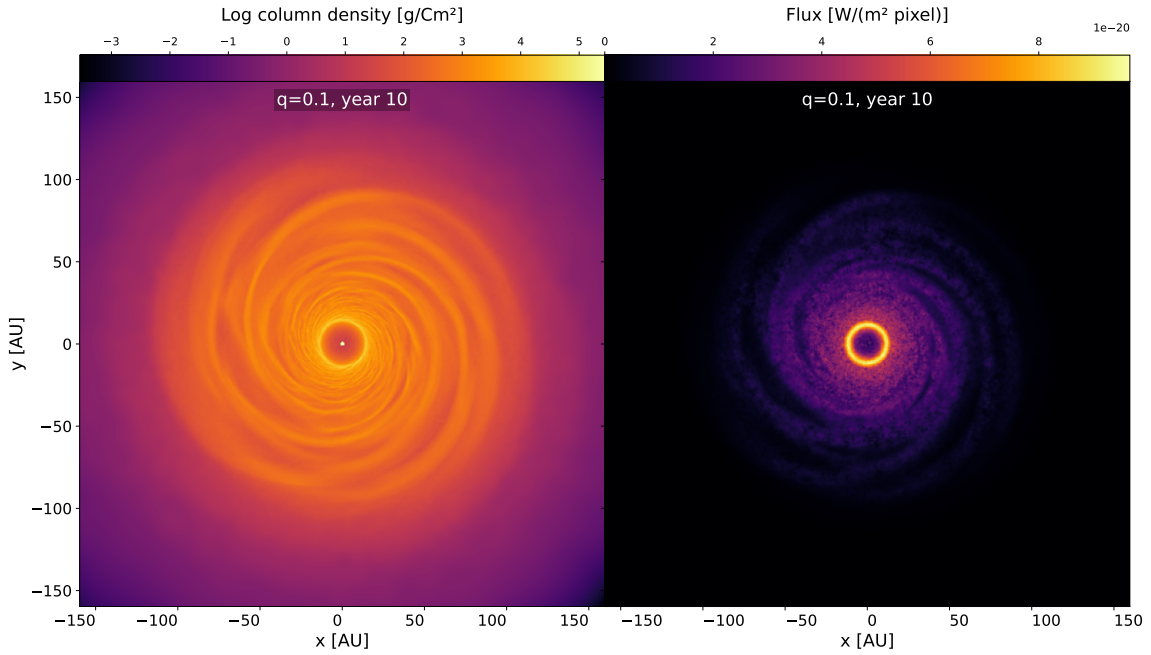


Figure 4.1: Snapshot of the hydrodynamical simulation (left panel) and corresponding image produced by MCFOST (right panel).

estimate of the Stokes number for such grains can indeed be provided:

$$\text{St} = \frac{\rho_d a}{\rho_g H_g} \approx \frac{\rho_d a}{\Sigma_g} = \frac{\rho_d a}{M_D} \pi R_D^2 = \frac{3 \frac{g}{cm^3} 0.1 cm}{0.1 M_\odot} \pi (100 \text{ AU})^2 \approx 0.011 \ll 1 \quad (4.1)$$

where for the disc mass and the grain size we've considered the least favourable combination, so for the other cases St is even smaller.

Consequently, the dust density distribution, which is proportional to the observed flux, exactly follows the gas density distribution in the midplane of the disc, allowing the tracking of GI.

Let's moreover note that a direct observation of gas emission wouldn't actually reproduce the correct structure of the disc: the gas is not fully deposited in the midplane and is not transparent to its own emission, thus the midplane emission is absorbed by higher layers and we could only observe the radiation coming from these parts of the disc.

4.2 Tracking the spiral arms and measuring velocity

The images produced were then analysed using some Python code that can be found at the following link: <https://github.com/mauroghira/thesis>. The given routine, described below, was first tested directly on the simulation snapshots³, where the spiral structure is evident; such a procedure allowed us to make a comparison between what is expected from the simulations and what would effectively be observed.

MCFOST outputs are `.fits` files that were read using the library `AstroPy` [Astropy Collaboration, 2013], [Astropy Collaboration, 2018].

To make the spiral structures more clear, the radial profile (the map of the azimuthal average of the image at each radius) was subtracted from the original image. Such an operation is justified because we've always assumed the disc to be axisymmetric in the absence of perturbations.

Figures 4.2 and 4.3 display the outputs of this workflow.

The tracking of the spiral arms was undertaken through the `scipy.ndimage.maximum_filter` method, utilised to spot the peaks of the observed flux and then isolate a single spiral arm through an iterative process which required active interaction with the user to choose the intervals of selection. In some cases, more than one spiral arm was tracked in the same image.

The tracks were saved as sets of (R, φ) coordinates so that the pattern velocity could be computed by comparing the value of the map $\varphi(R)$ at different times. Such datasets

³Actually, the first step (reading the data) was a bit different since the snapshots file type was different, and the second step (the subtraction of the radial profile) was not performed in this case.

CHAPTER 4. MEASURING PATTERN SPEED

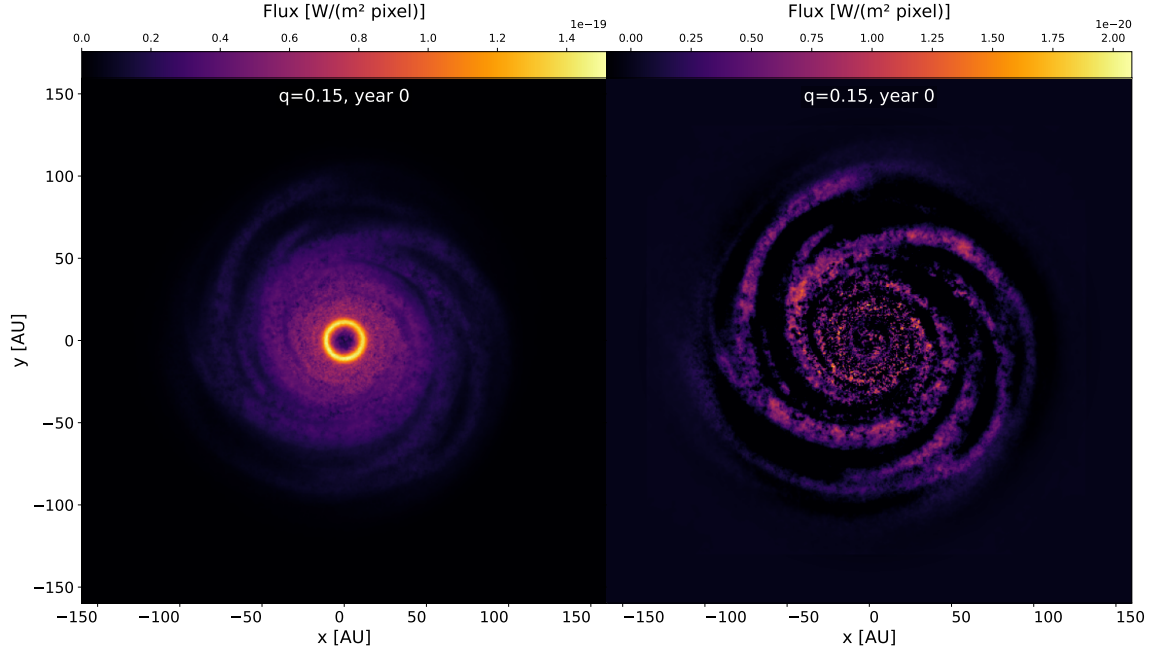


Figure 4.2: Comparison between the pure MCFOST image (left panel) and after having subtracted the radial map (right panel).

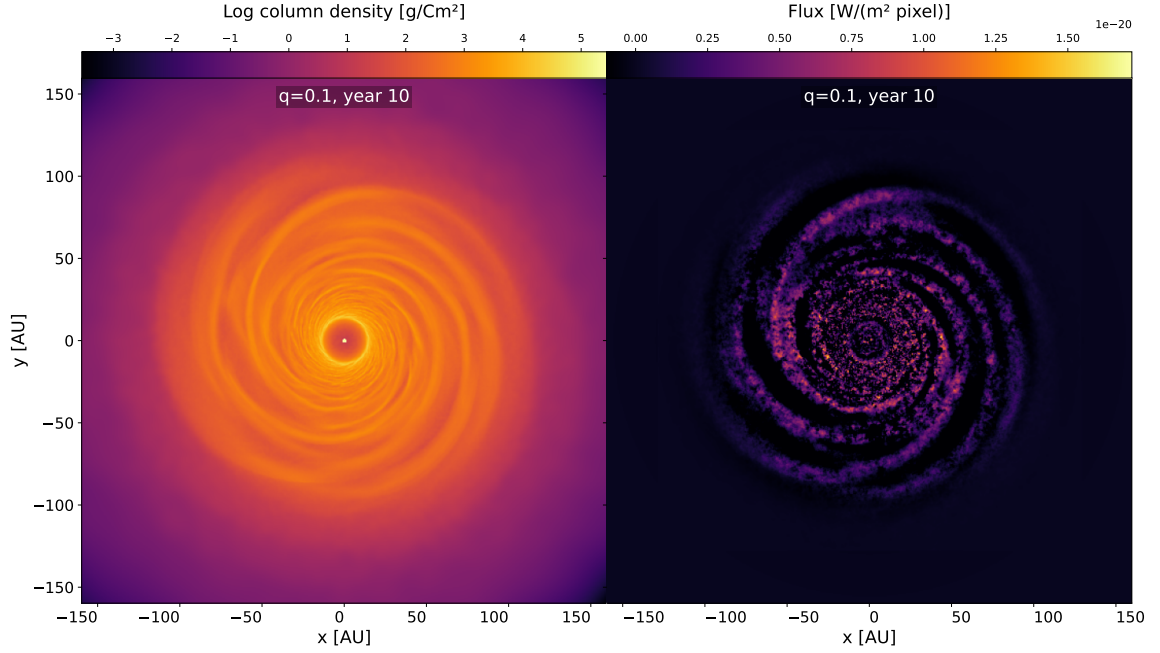


Figure 4.3: Comparison of the hydrodynamical simulation snapshot (left panel) with the MCFOST image after subtraction of the radial map (right panel).

CHAPTER 4. MEASURING PATTERN SPEED

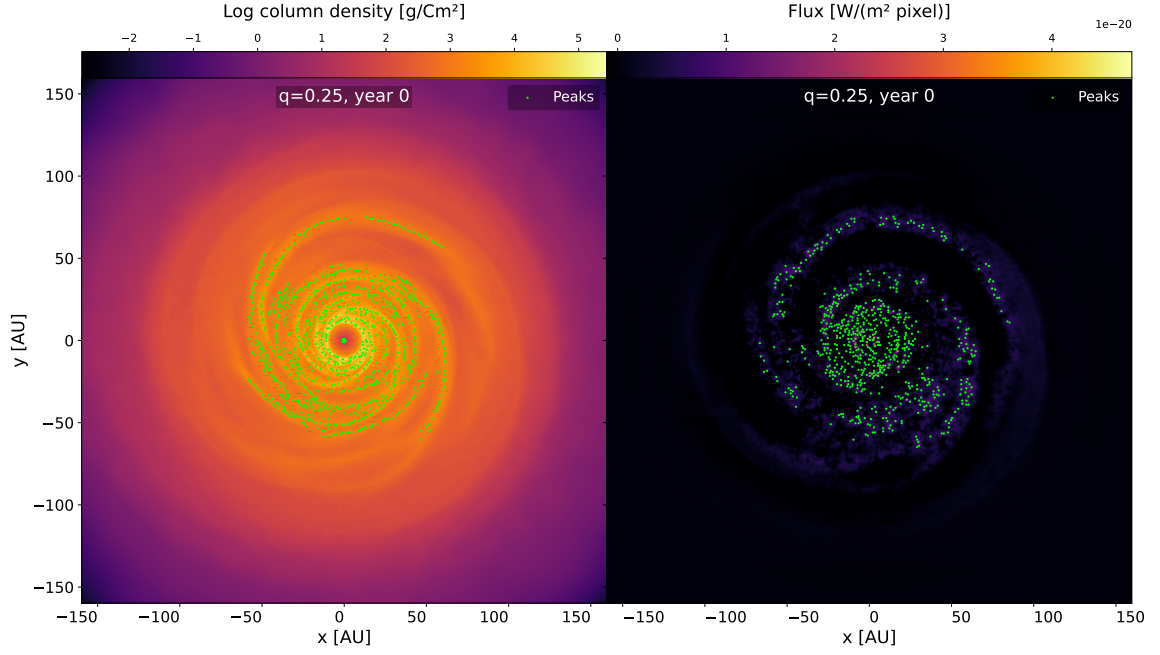


Figure 4.4: Spotted peaks in the simulation snapshots (left panel) and in the MCFOST image (right panel) at the first iteration.

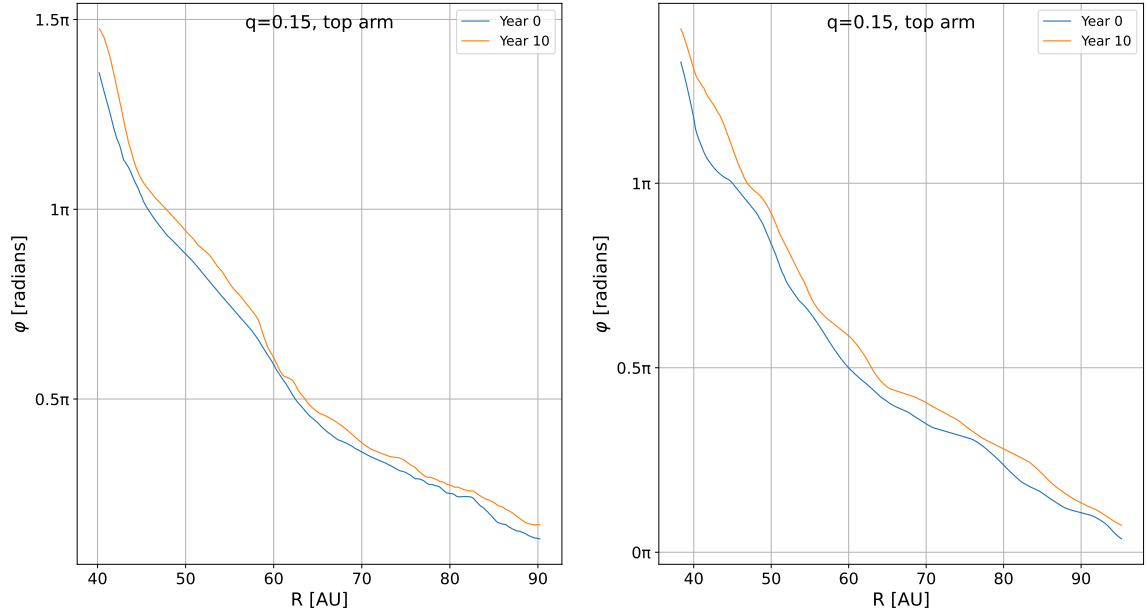


Figure 4.5: Comparison between the (R, φ) datasets obtained from a simulation (left) and the correspondent pseudo-observation (right).

CHAPTER 4. MEASURING PATTERN SPEED

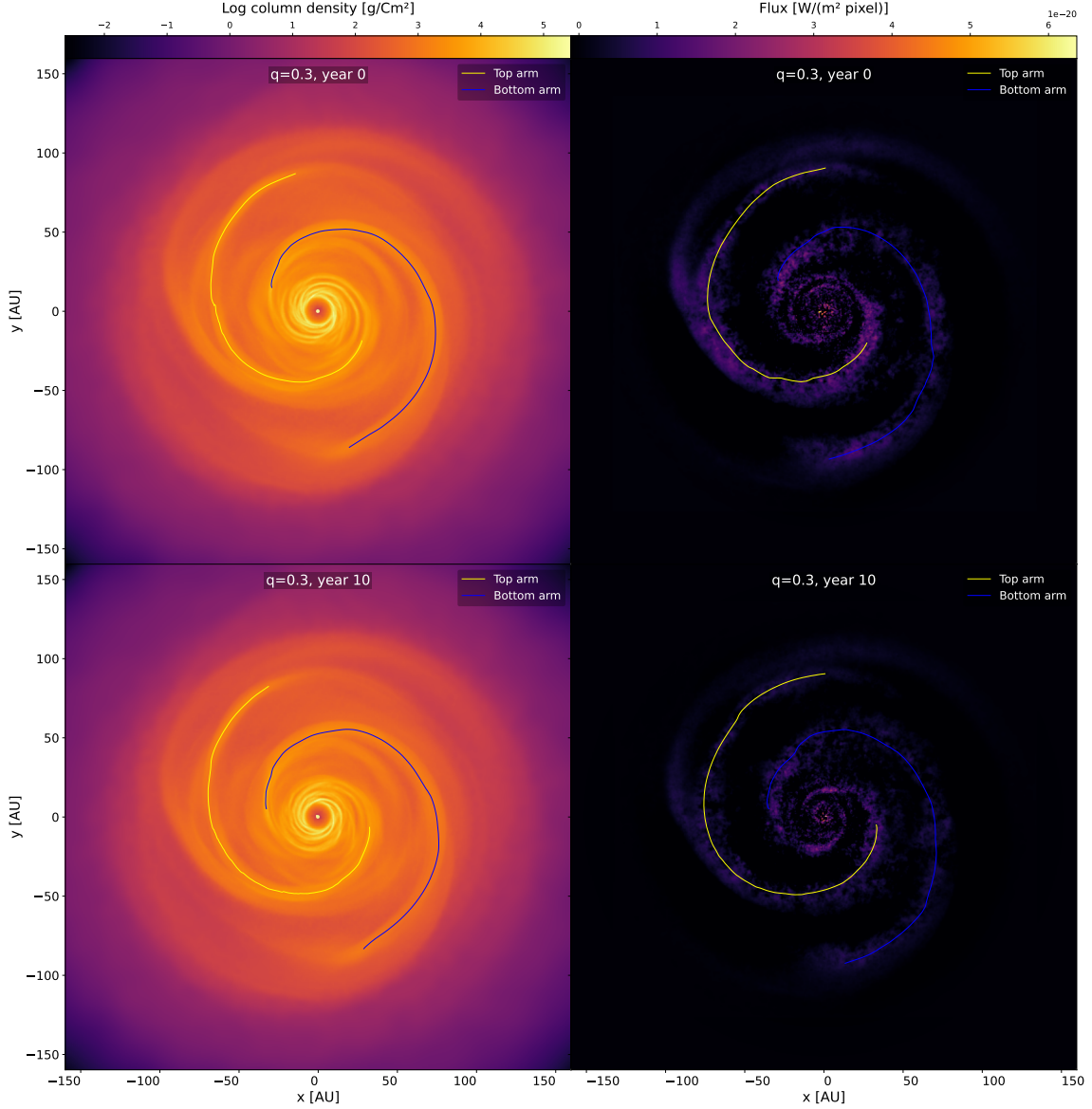


Figure 4.6: Detected spiral arms in the simulation snapshots (left panel) and in the MCFOST images (right panel).

were interpolated in a common range of radii⁴ using the `scipy.interpolate.interp1d` implementation, in order to ensure that the angle φ was evaluated at exactly the same points at different times. The outputs were moreover smoothed using the `uniform_filter1d` method from `scipy.ndimage`.

⁴The range was chosen taking the maximum value between the minima and the minimum between the maxima of the radial distances covered by the spiral arms at different times.

CHAPTER 4. MEASURING PATTERN SPEED

An example of the result of these steps is shown in Fig. 4.5 and 4.6.

Finally, the velocities obtained were fitted with a Keplerian-like function ($\Omega_p(R) = A \cdot R^{-3/2}$) and a constant trend ($\Omega_p(R) = B$), and the corresponding χ^2 were compared to check whether these measurements allow us to distinguish between GI and planetary spirals. For this purpose, the `scipy.optimize.curve_fit` method was exploited, and the χ^2 was estimated through the standard formula:

$$\chi^2 = \sum_{i=1}^N \left(\frac{y_i - f(x_i; \mathbf{p})}{\sigma_i} \right)^2 \quad \chi^2_\nu = \frac{\chi^2}{N - k} = \frac{\chi^2}{\nu}, \quad (4.2)$$

where y_i are the measured velocities, σ_i are their uncertainties, f is the model considered to fit the samples, $\mathbf{p}$ is the array of parameters of the model (having length k), N is the number of points, ν is the number of degrees of freedom, and χ^2_ν is the reduced chi-squared.

Using the implementation of the χ^2 distribution provided by the `scipy.stats.chi2` object, the associated confidence level (*p-value*) was also calculated. This parameter is simply the probability of obtaining a value of χ^2 greater than the one calculated under the assumption that the model is correct⁵ $P(\chi^2 \geq x | H_0)$. It also represents the probability of committing a first-type error, namely rejecting the model when it's correct.

As recalled by Eq. 4.2, to estimate the χ^2 , the uncertainties on the measured velocity were needed. Due to how the tracking process was carried out, the original datasets had the form $(R \pm \delta R, \varphi)$, thus the error on the radial distance had to be propagated through the operations described above.

These operations are non-linear and mix the original error in a non-trivial way; therefore, an analytical approach would have required keeping track of derivatives across all the transformations, which would have quickly become impractical. An alternative strategy involving a Monte Carlo simulation was therefore adopted: the input data were perturbed according to their given uncertainties, the full pipeline was repeated for many realizations, and the resulting scatter in the derived velocities was eventually computed. The above technique naturally takes into account the effects of interpolation, smoothing, and differencing without having to linearize them.

4.3 Results

As evident in Fig. 4.1 and 4.4, the images directly elaborated from the hydrodynamical simulations yielded significantly better-defined spiral arms, while even with the help

⁵More precisely, this parameter serves to test the acceptability of a certain hypothesis which, in our case, is: the model used to fit the data and the fitted parameters are correct, and the errors are independent and Gaussian.

CHAPTER 4. MEASURING PATTERN SPEED

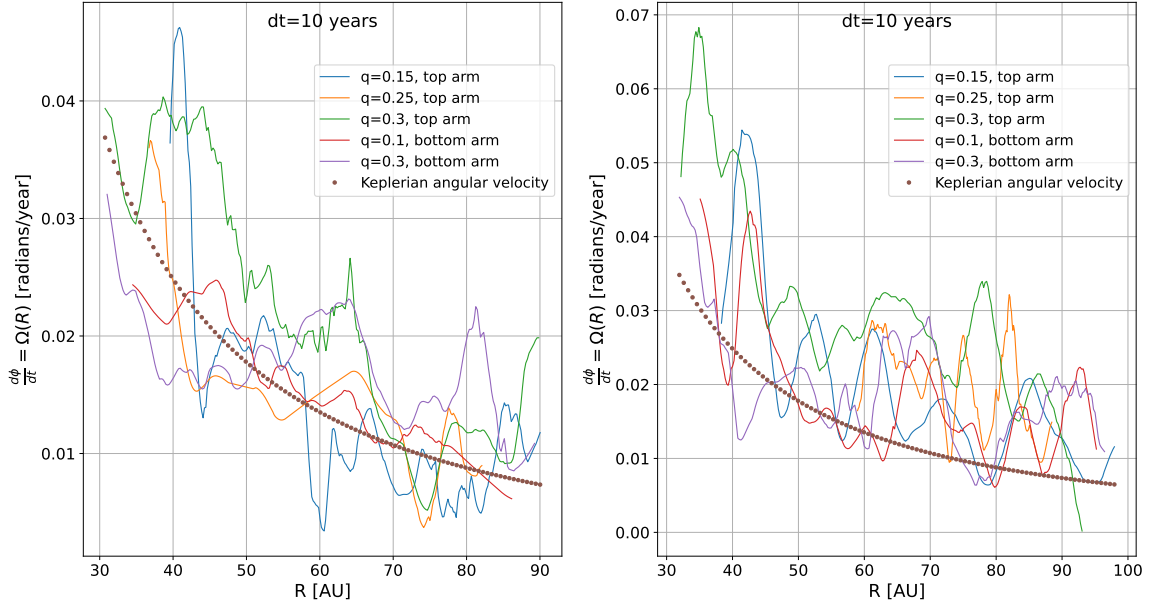


Figure 4.7: Pattern speeds computed from the simulations (left) and from the pseudo-observations (right).

of the radial map, the structures were far more blurred in the pseudo-observations. In particular, the peaks in the latter instance were much more scattered after the first iteration; hence, the tracking became considerably more difficult. Increasing the threshold on the intensity was useful to reduce the scattering, but finding a proper balance between the dispersion of the points and the possibility of tracking a spiral arm over a wide range of radii and angles was very challenging, given that in the outer regions of the disc the spirals were less prominent.

Let's note that, as expected, in low mass discs higher modes were excited more easily, making the tracking more intricate since secondary arms appeared, making the form of the principal one less recognizable.

Despite all this, in every case the pattern speed was found to be in the expected range, re-enforcing the prediction of co-rotating perturbations, as it can be observed in fig 4.7.

Let's moreover highlight the fact that the direct measurement allowed for an absolute estimation of Ω_p , going beyond theoretical predictions and numerical simulations' results, because they can only yield the absolute value of the difference $\Omega_p - \Omega_k$. In particular, the data revealed that $\Omega_p \gtrsim \Omega_k$.

The fits of the pattern speeds almost completely confirmed the better agreement of the measurements with the expected Keplerian behaviour, given that the reduced χ^2 are closer to unity for these fits. However, the associated p-values suggest a slightly more complex picture: in many cases, the confidence levels of the Keplerian fits are

CHAPTER 4. MEASURING PATTERN SPEED

very close to 100%, since the measured reduced chi-squared are quite lower than unity; such a situation is still statistically acceptable, but looks suspicious and is likely due to the process of smoothing which reduces the variability of the data making the uncertainties look overestimated. In spite of that, such fits remain more acceptable than the corresponding constant ones, because the p-values of the latter are lower than 1%.

Conversely, the bottom arm in the disc with $q = 0.3$ and the top arm in the disc with $q = 0.25$ stand out of the framework exposed above: the pattern speeds here seem indeed to be more compatible with the constant trend. However, from the graph reported in Fig. 4.7 it is actually evident that there the measured velocity fluctuated a lot, so both the fits are not reliable, expectedly for the data of the MCFOST image. Moreover, in light of the other results, these discrepancies can be fully ascribed to the imprecision of the tracking.

Fig. 4.8 shows an example of the fits performed, while all the reduced χ^2 obtained, along with the corresponding confidence levels, are reported in Tab. 4.2

As said in the previous section, for the discs with $q = 0.15$ and 0.3 , it was attempted to make a more precise measurement by tracking the top spiral arm at the fifth year of simulation. Unfortunately, the precision of the tracking process was not enough to provide meaningful results from the pseudo-observations, while a certain degree of agreement was found when working directly on the simulations. In the former case, indeed, the great dispersion made it quite arduous to insert a third dataset of (R, φ) coordinates between the two already available, since they are already quite close, as it can be seen from Fig. 4.5. In Fig. 4.9 and 4.10 the outcomes of this attempt are displayed.

CHAPTER 4. MEASURING PATTERN SPEED

Mass ratio	Arm		Keplerian Fit		Constant Fit	
			χ_ν	p	χ_ν	p
0.1	Bottom	Simulation	0.707	100%	6.730	<1%
		Observation	0.977	59%	1.411	<1%
0.15	Top	Simulation	0.828	98%	1.702	<1%
		Observation	0.778	100%	1.459	<1%
0.25	Top	Simulation	0.829	98%	1.066	20%
		Observation	0.786	100%	1.130	7%
0.3	Bottom	Simulation	1.450	<1%	0.612	100%
		Observation	1.057	13%	0.957	70%
	Top	Simulation	0.733	99%	3.875	<1%
		Observation	0.928	72%	2.110	<1%

Table 4.2: Results of the fits on the measured velocities.

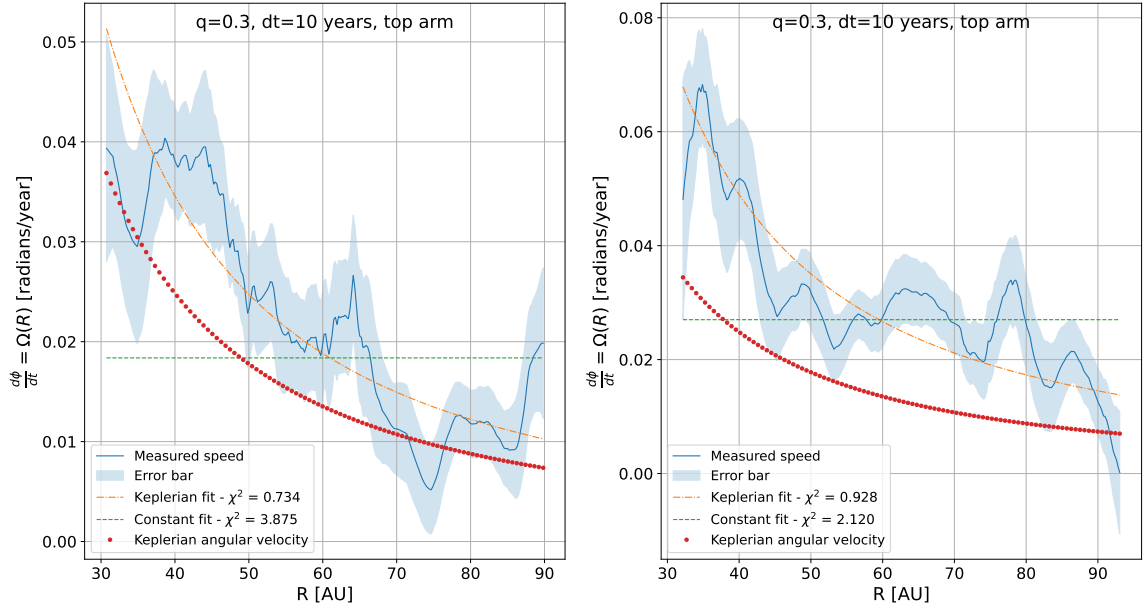


Figure 4.8: Computed pattern speed and corresponding uncertainty compared with the fitted data from a simulation (left) and the associated MCFOST image (right).

CHAPTER 4. MEASURING PATTERN SPEED

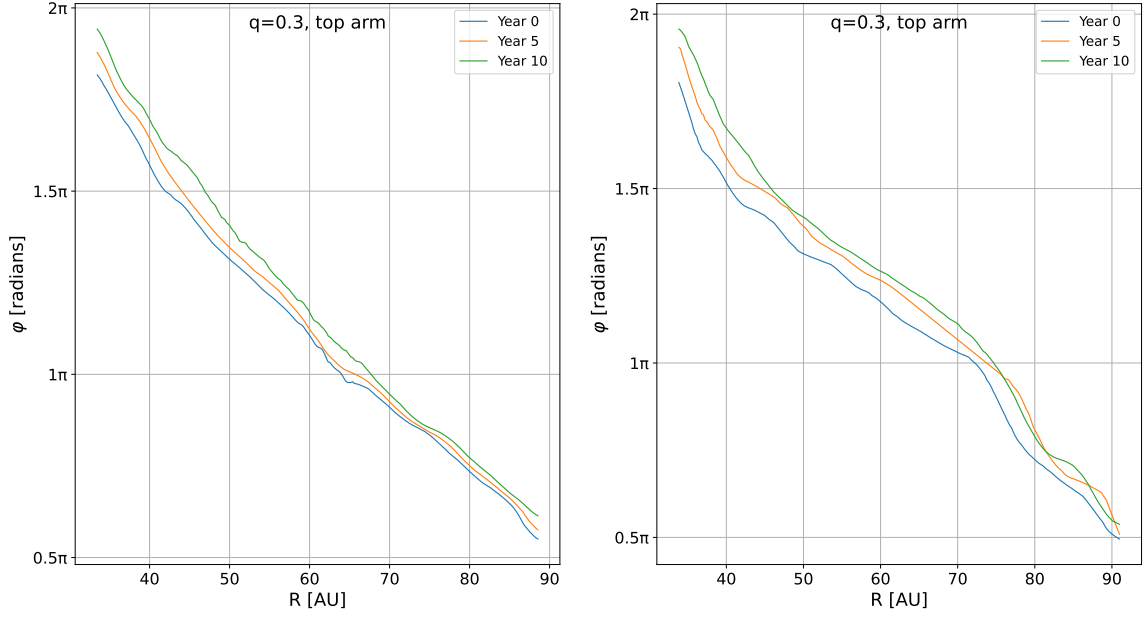


Figure 4.9: Comparison between the (R, ϕ) datasets obtained from a simulation (left) and the correspondent pseudo-observation (right).

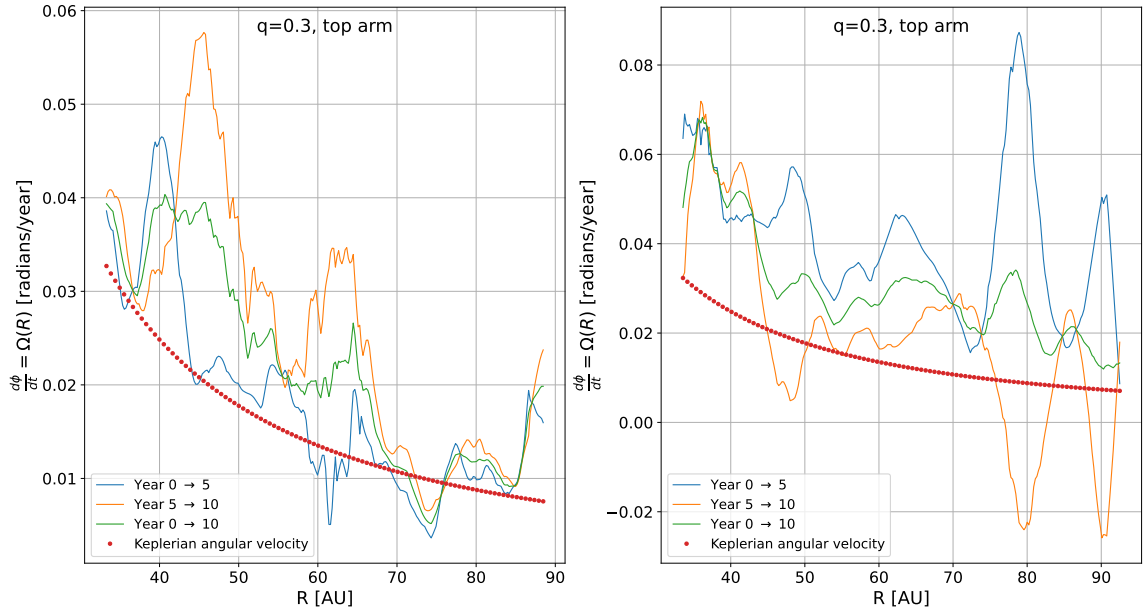


Figure 4.10: Pattern speeds obtained from a simulation (left) and the correspondent pseudo-observation (right).

Chapter 5

Conclusions

From the analysis of the development of gravitational instabilities in the WKB approximation, the dispersion relation

$$(\omega - m\Omega)^2 = c_s^2 k^2 - 2\pi G \Sigma |k| + \kappa^2 \quad (5.1)$$

was derived. Due to thermal balance, discs are expected to lie in a marginally stable condition, where only the most unstable modes ($k \approx k_0 = \frac{\pi G \Sigma}{c_s^2}$) can be excited. These perturbations are thus expected to co-rotate with the disc, as entailed by Eq. 5.1.

This prediction has been confirmed by a Fourier analysis of the excited modes in hydrodynamical simulations [Cossins et al., 2009], from which the wavenumbers of the perturbation were measured and then substituted back in Eq. 5.1.

The analysis of pseudo-observations of discs carried out in this thesis further corroborated these predictions and added some relevant information about the exact value of the pattern speed.

We found that 10 years is a suitable time span to compute pattern speed from direct observations; the precision of this measurement is highly dependent on the accuracy of the tracking procedure, which is very laborious even in the hydrodynamical simulations since the spiral structures are highly dynamic and chaotic. Anyway, the obtained velocities all lay in the expected range.

Evidences in favour of a radius-dependent profile compatible with a Keplerian-like trend were found, so we can conclude that it is indeed possible to distinguish between GI spirals and planetary ones from observations. We also established that performing more precise calculations by decreasing the time span is, in principle, possible, as a compatible measurement of the pattern speed was obtained from the direct analysis of the simulations snapshots every 5 years. However, the accuracy of the tracking method cannot fully balance the greater scattering of the data from the pseudo-observations.

In light of this, the main challenge for the future is improving the tracking pipeline:

CHAPTER 5. CONCLUSIONS

to do so, other implementations (e.g. the ones from `scipy.spatial.KDTree` class) may be combined. Moreover, a machine learning-based approach might make the process more automatic, but to train such an algorithm, a large amount of already tracked spirals would be needed. The chain constructed for this thesis may indeed be a starting point to collect sufficient data to realize such an implementation.

Finally, it would be interesting to investigate whether the same results can be obtained in a more realistic and general case of a tilted (not face-on) disc. In this case, an extra step would have to be inserted at the beginning of our routine to de-project the image.

Appendix A

Derivation of hydrodynamics equations

A.1 Derivative operations in cylindrical coordinates

Here's a brief recap of the relevant expressions for our calculations:

- Scalar fields:

$$\nabla \rho = \frac{\partial \rho}{\partial R} \hat{R} + \frac{1}{R} \frac{\partial \rho}{\partial \varphi} \hat{\varphi} + \frac{\partial \rho}{\partial z} \hat{z} \quad (\text{A.1})$$

- Vector fields:

$$\nabla \cdot \mathbf{A} = \frac{1}{R} \frac{\partial}{\partial R} (R A_R) + \frac{1}{R} \frac{\partial A_\varphi}{\partial \varphi} + \frac{\partial A_z}{\partial z} \quad (\text{A.2})$$

$$\nabla \mathbf{A} = \begin{pmatrix} \frac{\partial A_R}{\partial R} & \frac{\partial A_R}{R \partial \varphi} - \frac{A_\varphi}{R} & \frac{\partial A_R}{\partial z} \\ \frac{\partial A_\varphi}{\partial R} & \frac{\partial A_\varphi}{R \partial \varphi} + \frac{A_R}{R} & \frac{\partial A_\varphi}{\partial z} \\ \frac{\partial A_z}{\partial R} & \frac{\partial A_z}{R \partial \varphi} & \frac{\partial A_z}{\partial z} \end{pmatrix} \quad (\text{A.3})$$

- Tensor fields:

$$\nabla \cdot \mathbf{T} = \begin{pmatrix} \frac{\partial T_{rr}}{\partial R} + \frac{\partial T_{R\varphi}}{R \partial \varphi} + \frac{T_{rr} - T_{\varphi\varphi}}{R} + \frac{\partial T_{rz}}{\partial z} \\ \frac{\partial T_{\varphi R}}{\partial R} + \frac{\partial T_{\varphi\varphi}}{R \partial \varphi} + \frac{T_{R\varphi} + T_{\varphi R}}{R} + \frac{\partial T_{\varphi z}}{\partial z} \\ \frac{\partial T_{zr}}{\partial R} + \frac{\partial T_{z\varphi}}{R \partial \varphi} + \frac{T_{zr}}{R} + \frac{\partial T_{zz}}{\partial z} \end{pmatrix} \quad (\text{A.4})$$

A.2 Vertically integrated continuity equation

Let's start from Eq. 2.2 and exploit A.2; since in the axisymmetric disc approximation $\frac{\partial}{\partial \varphi} = 0$, we obtain:

$$\nabla \cdot (\rho \mathbf{v}) = \frac{1}{R} \frac{\partial}{\partial R} (R \rho v_R) + \frac{\partial (R \rho v_z)}{\partial z}. \quad (\text{A.5})$$

In the thin disc approximation, $H \ll R$, $\lim_{z \rightarrow \infty} \rho = 0$, and v_z is negligibly small, so

$$\int_{-\infty}^{\infty} dz \frac{\partial}{\partial z} (R \rho v_z) = R \int_{-\infty}^{\infty} dz \frac{\partial}{\partial z} (\rho v_z) = R (\rho v_z)|_{-\infty}^{\infty} = 0.$$

Moreover, we can pass the integral inside the radial derivative, since they act on different coordinates, so the first term obtained above exactly becomes

$$\frac{1}{R} \frac{\partial}{\partial R} \left[R \int_{-\infty}^{\infty} dz \rho v_R \right] = \frac{1}{R} \frac{\partial}{\partial R} \left[R \Sigma \left(\frac{1}{\Sigma} \int_{-\infty}^{\infty} dz \rho v_R \right) \right] = \frac{1}{R} \frac{\partial}{\partial R} (R \Sigma \tilde{v}_R),$$

where we've introduced the vertically averaged radial velocity $\tilde{v}_R \doteq \frac{1}{\Sigma} \int_{-\infty}^{\infty} dz \rho v_R$.

Therefore we've exactly recovered Eq. 2.7

$$\frac{\partial \Sigma}{\partial t} + \frac{1}{R} \frac{\partial}{\partial R} (R \Sigma \tilde{v}_R) = 0 \quad (\text{A.6})$$

A.3 Vertically integrated Navier-Stokes equations

Let's now focus on Eq. 2.1, giving some insights on every component:

A.3.1 Vertical component

This component of the equations is quite simple since it can be written as

$$\rho \left(\frac{\partial v_z}{\partial t} + v_R \frac{\partial v_z}{\partial R} + \frac{v_\varphi}{R} \frac{\partial v_z}{\partial \varphi} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial P}{\partial z} - \rho \frac{\partial \Phi}{\partial z} + (\nabla \cdot \sigma)_z. \quad (\text{A.7})$$

The last term on the right-hand side vanishes because the only non-zero term in the stress tensor is the $R\varphi$ entry, which doesn't appear in the expression for the divergence given above (A.4). Moreover, the thin disc approximation allows us to neglect the whole left-hand side of the equation, since v_z is very small.

Thus, we are left with the hydrostatic balance equation:

$$\frac{1}{\rho} \frac{\partial P}{\partial z} = -\frac{\partial \Phi}{\partial z}, \quad (\text{A.8})$$

which we can specify in some particular cases as done in sect. 2.1:

APPENDIX A. DERIVATION OF HYDRODYNAMICS EQUATIONS

- **Keplerian disc**¹: $\Phi = -\frac{GM_*}{r} \Rightarrow$

$$\frac{\partial \Phi}{\partial z} = \frac{GM_*}{r^2} \frac{\partial r}{\partial z} = \frac{GM_*}{r^3} z \approx \frac{GM_*}{R^3} z \quad \Rightarrow \quad \frac{c_s^2}{\rho} \frac{\partial \rho}{\partial z} = -\frac{GM_*}{R^3} z. \quad (\text{A.9})$$

The solution of this differential equation in the case of c_s independent from z is simply

$$\rho(z) = \rho_0 \exp\left(-\frac{z^2}{2H_{nsg}^2}\right), \quad (\text{A.10})$$

where $H_{nsg} \doteq \frac{c_s}{\Omega_K}$ defines the characteristic thickness of the system.

- **Self-gravitating disc**: solving the Poisson equation defining Φ in this case strongly depends on the model assumed; for an infinite, thin, radially homogeneous ($\Sigma(R) = \Sigma_0$) disc, we obtain

$$\frac{\partial \Phi}{\partial z} = 2\pi G \Sigma_0 = \frac{c_s^2}{\rho} \frac{\partial \rho}{\partial z}. \quad (\text{A.11})$$

This equation has an analytical solution in the case of constant sound speed, namely

$$\rho(z) = \rho_0 \frac{1}{\cosh^2(z/H_{sg})}, \quad (\text{A.12})$$

where, once again, $H_{sg} \doteq \frac{c_s^2}{\pi G \Sigma_0}$ defines the characteristic thickness. This expression can be used to define the characteristic thickness in general, considering the specific expression for $\Sigma(R)$.

- **Mixed systems**: if the gravitational potential is made up of both the components, we can evaluate which contribution dominates by comparing the expressions for $\frac{\partial \Phi}{\partial z}$ obtained above, thus recovering the relation

$$\frac{GM_* H_{nsg}}{R^3} \approx 2\pi G \Sigma_0 \quad \Rightarrow \quad \frac{M_D(R)}{M_*} \approx \frac{H_{nsg}}{R} \quad (\text{A.13})$$

since a rough estimation of the mass of the disc within a given radius R is given by $M_D(R) \approx \pi R^2 \Sigma$ (the correct value would be $M_D(R) = \int_0^{2\pi} d\varphi \int_0^R dx \Sigma(x, \varphi)$).

¹ $r = \sqrt{R^2 + z^2}$ is the radial distance in spherical coordinates. In the thin disc case, $r \approx R$.

A.3.2 Radial component

Using again the rules A.3 and A.4, this component becomes

$$\rho \left[\frac{\partial v_R}{\partial t} + v_R \frac{\partial v_R}{\partial R} + v_\varphi \left(\frac{1}{R} \frac{\partial v_R}{\partial \varphi} - \frac{v_\varphi}{R} \right) + v_z \frac{\partial v_R}{\partial z} \right] = - \frac{\partial P}{\partial R} + \frac{\partial \sigma_{RR}}{\partial R} + \frac{1}{R} \frac{\partial \sigma_{R\varphi}}{\partial \varphi} + \frac{\partial \sigma_{Rz}}{\partial z} + \frac{1}{R} (\sigma_{RR} - \sigma_{\varphi\varphi}) - \rho \frac{\partial \Phi}{\partial R}. \quad (\text{A.14})$$

Exploiting once again the various approximations:

- axisymmetric disc $\Rightarrow \frac{\partial}{\partial \varphi} = 0$;
- classical shear viscosity $\Rightarrow \sigma_{lj} = \sigma_{ji} \neq 0 \Leftrightarrow i = \varphi, j = R$;
- thin disc $\Rightarrow H/R \ll 1, v_z \sim 0$;
- $v_R \ll c_s \ll v_\varphi \Rightarrow \frac{\partial v_R}{\partial t}$ negligible;
- barotropic gas $\Leftrightarrow P = P(\rho)$ only $\Rightarrow c_s^2 \doteq \frac{dP}{d\rho}$;²

we get

$$-\rho \frac{v_\varphi^2}{R} = -\frac{\partial P}{\partial R} - \rho \frac{\partial \Phi}{\partial R} \Rightarrow \frac{v_\varphi^2}{R} = \frac{c_s^2}{\rho} \frac{\partial \rho}{\partial R} + \frac{\partial \Phi}{\partial R}. \quad (\text{A.15})$$

Once again, we can specify this equation in some particular cases:

- **Keplerian disc:**

$$\begin{aligned} \Phi \approx -\frac{GM_*}{R} &\Rightarrow v_\varphi = v_k \sqrt{1 + \left(\frac{c_s}{v_k} \right)^2 \frac{\partial \ln \rho}{\partial \ln R}} \approx v_k = \sqrt{\frac{GM}{R}} \\ &\Rightarrow \Omega = \frac{v_\varphi}{R} \approx \Omega_k = \sqrt{\frac{GM}{R^3}}, \end{aligned} \quad (\text{A.16})$$

where the last approximation holds since $\frac{c_s}{v_\varphi} = \frac{H}{R} \ll 1$ and the density gradients are usually small.

- **Self-gravitating disc:** as before, a simple expression for the gravitational potential can be found in some particular cases; adopting the *Mestel model*, in which the disc is assumed to extend to infinity with a density distribution $\Sigma(R) = \Sigma_0 \frac{R_0}{R}$, we are left with (we directly neglect the density gradient term in the equation as justified above)

$$\begin{aligned} \frac{\partial \Phi}{\partial R} = 2\pi G \Sigma(R) &\Rightarrow v_\varphi^2 = 2\pi G \Sigma R = 2\pi G \Sigma_0 R_0 = \text{const} \\ &\Rightarrow \Omega = \frac{\sqrt{2\pi G \Sigma_0 R_0}}{R} \end{aligned} \quad (\text{A.17})$$

²This definition can be justified by looking for the wave equation for sound waves, which are density (or equivalently pressure) waves; this equation can be obtained by considering the linearized continuity and Euler equation in a perturbative approach like the one adopted in sect. 2.2.1.

APPENDIX A. DERIVATION OF HYDRODYNAMICS EQUATIONS

- **Mixed systems:** if the gravitational potential is made up of both the components, we can evaluate which contribution dominates by comparing the expressions for $\frac{\partial \Phi}{\partial R}$ obtained above, thus recovering the expected relation

$$\frac{GM_*}{R^2} \approx 2\pi G\Sigma \Rightarrow M_* = 2\pi R^2 \Sigma(R) \approx M_D(R) . \quad (\text{A.18})$$

A.3.3 Azimuthal component

Using Eq. A.3 and A.4 and integrating in the vertical direction, the angular component of the Navier-Stokes equations becomes

$$\rho \left[\frac{\partial v_\varphi}{\partial t} + v_R \frac{\partial v_\varphi}{\partial R} + v_\varphi \left(\frac{1}{R} \frac{\partial v_\varphi}{\partial \varphi} + \frac{v_R}{R} \right) + v_z \frac{\partial v_\varphi}{\partial z} \right] = - \frac{\partial P}{\partial \varphi} - \rho \frac{\partial \Phi}{\partial \varphi} + \frac{\partial \sigma_{\varphi R}}{\partial R} + \frac{1}{R} \frac{\partial \sigma_{\varphi \varphi}}{\partial \varphi} + \frac{\partial \sigma_{\varphi z}}{\partial z} + \frac{1}{R} (\sigma_{\varphi R} + \sigma_{\varphi \varphi}) ; \quad (\text{A.19})$$

however, using once again the axisymmetric approximation ($\frac{\partial}{\partial \varphi} = 0$), the thin disc approximation ($v_z \sim 0$), and the classic shear viscosity stress tensor ($\sigma_{lj} = \sigma_{ji} \neq 0 \Leftrightarrow i = \varphi, j = R$), we are left with

$$\rho \left[\frac{\partial v_\varphi}{\partial t} + v_R \frac{\partial v_\varphi}{\partial R} + \frac{v_R v_\varphi}{R} \right] = \frac{\partial \sigma_{R\varphi}}{\partial R} + \frac{2}{R} \sigma_{R\varphi} ,$$

which integrated in the vertical direction assuming the velocities do not depend on z due to the thin disc approximation becomes

$$R\Sigma \left[\frac{\partial v_\varphi}{\partial t} + v_R \frac{\partial v_\varphi}{\partial R} \right] = R \frac{\partial T_{R\varphi}}{\partial R} + 2T_{R\varphi} - \Sigma v_R v_\varphi . . \quad (\text{A.20})$$

Let's now analyse Eq. 2.10 term by term to show the equivalence; by simply using the Leibniz rule we get:

$$\begin{aligned} \bullet \quad \frac{\partial}{\partial t} (\Sigma R v_\varphi) &= R \frac{\partial \Sigma}{\partial t} v_\varphi + R \Sigma \frac{\partial v_\varphi}{\partial t} \stackrel{(2.7)}{=} R \Sigma \frac{\partial v_\varphi}{\partial t} - v_\varphi \frac{\partial}{\partial R} (\Sigma R v_R) \\ &= R \Sigma \frac{\partial v_\varphi}{\partial t} - R v_\varphi \frac{\partial}{\partial R} (\Sigma v_R) - v_\varphi \Sigma v_R ; \\ \bullet \quad \frac{1}{R} \frac{\partial}{\partial R} (R^2 v_\varphi v_R \Sigma) &= 2 v_\varphi v_R \Sigma + R \frac{\partial}{\partial R} (v_\varphi v_R \Sigma) ; \end{aligned}$$

and putting everything together we obtain

$$\begin{aligned} R \Sigma \frac{\partial v_\varphi}{\partial t} + R \frac{\partial}{\partial R} (v_\varphi v_R \Sigma) - R v_\varphi \frac{\partial}{\partial R} (\Sigma v_R) + \Sigma v_\varphi v_R &= R \Sigma \frac{\partial v_\varphi}{\partial t} + R \Sigma v_R \frac{\partial v_\varphi}{\partial R} + \Sigma v_\varphi v_R \\ &= R \Sigma \left(\frac{\partial v_\varphi}{\partial t} + v_R \frac{\partial v_\varphi}{\partial R} \right) + \Sigma v_\varphi v_R \stackrel{(A.20)}{=} R \frac{\partial T_{R\varphi}}{\partial R} + 2T_{R\varphi} \quad (\text{A.21}) \end{aligned}$$

APPENDIX A. DERIVATION OF HYDRODYNAMICS EQUATIONS

Remembering that $T_{R\varphi} = v\Sigma R \frac{\partial \Omega}{\partial R}$, we finally can rewrite the right-hand term in Eq. 2.10 as

$$\frac{1}{R} \frac{\partial}{\partial R} \left(v\Sigma R^3 \frac{\partial \Omega}{\partial R} \right) = \frac{1}{R} \frac{\partial}{\partial R} (R^2 T_{R\varphi}) = \frac{1}{R} 2RT_{e_\varphi} + \frac{1}{R} R^2 \frac{\partial T_{R\varphi}}{\partial R}, \quad (\text{A.22})$$

which is exactly equal to the result in Eq. A.21.

Appendix B

Generalities on radiative transport

Let's recall some definitions in the theory of radiative transport we've used in the previous chapters and some properties employed by the MCFOST code.

B.1 Fundamental definitions

First of all, let's note that every quantity we will introduce can be taken at a specific frequency or integrated over a range: $X = \int_{\nu_1}^{\nu_2} X_\nu d\nu$.

The fundamental property of a source of radiation is its *luminosity* $\mathcal{L} \doteq \frac{dE}{dt}$, the amount of energy irradiated per unit time. If we are interested in quantifying the energy transported through a region of area dA , we should then introduce the *flux* $F \doteq \frac{dE}{dt dA}$ and the *intensity* (or radiance)

$$I \doteq \frac{dE}{dt dA d\Omega}, \quad (\text{B.1})$$

which takes into account the different directions of the radiation crossing the surface element ($d\Omega$ is the unit solid angle). It is straightforward to see that the relation between these two quantities is simply

$$F = \int_{\Omega} I(\vartheta, \varphi) \cos \vartheta d\Omega, \quad (\text{B.2})$$

where the $\cos \vartheta$ factor comes from considering the effective area "seen" by a beam that is not normal to the given surface.

When considering radiation passing through a medium, we should moreover bring in some other essential quantities¹:

¹I've defined them per unit frequency to highlight the fact that the emission and absorption properties of a medium can vary a lot with the frequency of the radiation.

APPENDIX B. GENERALITIES ON RADIATIVE TRANSPORT

1. The *emissivity* (or spontaneous emission coefficient) $j_\nu \doteq \frac{dE}{dt dV d\Omega d\nu}$, representing the energy emitted per unit time, solid angle and volume by the medium.
2. The *absorption coefficient* α_ν , representing the fraction of radiation absorbed per unit length; this parameter can be defined as the inverse of the mean free path of the photons in the medium or, equivalently, as the product of the numerical density of the particles in the medium and their absorption cross-section (σ_ν), since it only depends on the microscopic features of the system. Furthermore, α_ν can be related to the *opacity* of the medium, which is precisely defined as

$$\kappa_\nu \doteq \frac{\alpha_\nu}{\rho} = \frac{\sigma_\nu}{m} , \quad (\text{B.3})$$

where m is the mean mass of the particles in the medium.

3. The *optical depth* $d\tau_\nu \doteq \alpha_\nu ds$; we've implicitly used this parameter in Chapter 2 when talking about optically thin or thick discs.
4. The *source function* $S_\nu \doteq \frac{j_\nu}{\alpha_\nu}$.

We can now write the fundamental equation of radiative transport, which regulates the evolution of the intensity of a beam travelling along the direction s in a medium:

$$\frac{dI_\nu}{ds} = -\alpha_\nu I_\nu + j_\nu \Leftrightarrow \frac{dI_\nu}{d\tau_\nu} = -I_\nu + S_\nu . \quad (\text{B.4})$$

Its general solution is

$$I_\nu(\tau_\nu) = I_\nu(0) e^{-\tau_\nu} + \int_0^{\tau_\nu} S_\nu(t) e^{-(\tau_\nu-t)} dt , \quad (\text{B.5})$$

where the first term represents the progressive absorption of the initial intensity and the second one the intensity of the radiation emitted at a depth τ_ν , which is in turn absorbed as it propagates in the medium.

Having found this solution, we can now make some considerations about the thickness of the medium and its consequences on the radiation eventually emitted by the system.

B.2 Thermal emission

We call *thermal emission* the radiation coming from a body at thermal equilibrium at a given temperature T . A very special case of this kind of emission is the blackbody radiation, in which the emitting object is at thermal equilibrium with the radiation

APPENDIX B. GENERALITIES ON RADIATIVE TRANSPORT

too; equivalently, this condition can be expressed in terms of the optical depth, which has to be infinite: in this special case, indeed, a photon entering the region considered is completely absorbed, thus the only radiation eventually emitted is the one due to the spontaneous emission processes occurring at the temperature of the object.

An interesting feature of thermal emission is that, even in the case of finite optical depth, the source function is always the Planckian

$$B_\nu = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1} . \quad (\text{B.6})$$

This can be justified by noting that the source function is an intrinsic property of the body itself, since at thermal equilibrium the microscopic processes considered to express j_ν and α_ν are fully determined by the temperature of the system, which is the only parameter affecting the population of the energetic levels; thus it is independent from the radiation field the object may be immersed in. If we put the body in a cavity filled with blackbody radiation at a given temperature ($I_\nu = B_\nu(T)$), it will come into equilibrium at the same temperature and since the blackbody radiation field is isotropic ($\frac{dI_\nu}{ds} = 0$), Eq. B.4 reduces exactly to $S_\nu = B_\nu(T)$. As a consequence, the source function must be the Planckian in all the cases of thermal equilibrium.

Let's now briefly recall the main properties of blackbody emission:

1. **Stefan-Boltzmann law:** $F = \sigma_{SB} T^4$; when observing a system, an useful quantity defined considering this law is the *Effective temperature*, the temperature the system should have to generate a certain observed flux if its emission were blackbody. We've already used this definition in sect. 2.3 to compute the temperature profile of an active disc.
2. **Wien displacement law:** $T \lambda_{max} = \text{const.}$
3. **Rayleigh-Jeans approximation:**

$$h\nu \ll k_B T \Rightarrow B_\nu(t) = \frac{2\nu^2}{c^2} k_B T .$$

4. **Wien approximation:**

$$h\nu \gg k_B T \Rightarrow B_\nu(T) = \frac{2h\nu^3}{c^2} e^{-h\nu/k_B T} .$$

In Fig. B.1 it was shown how these approximations still hold true for an active disc SED.

5. **Brightness temperature:** the temperature a blackbody should have to generate the observed radiance.

APPENDIX B. GENERALITIES ON RADIATIVE TRANSPORT

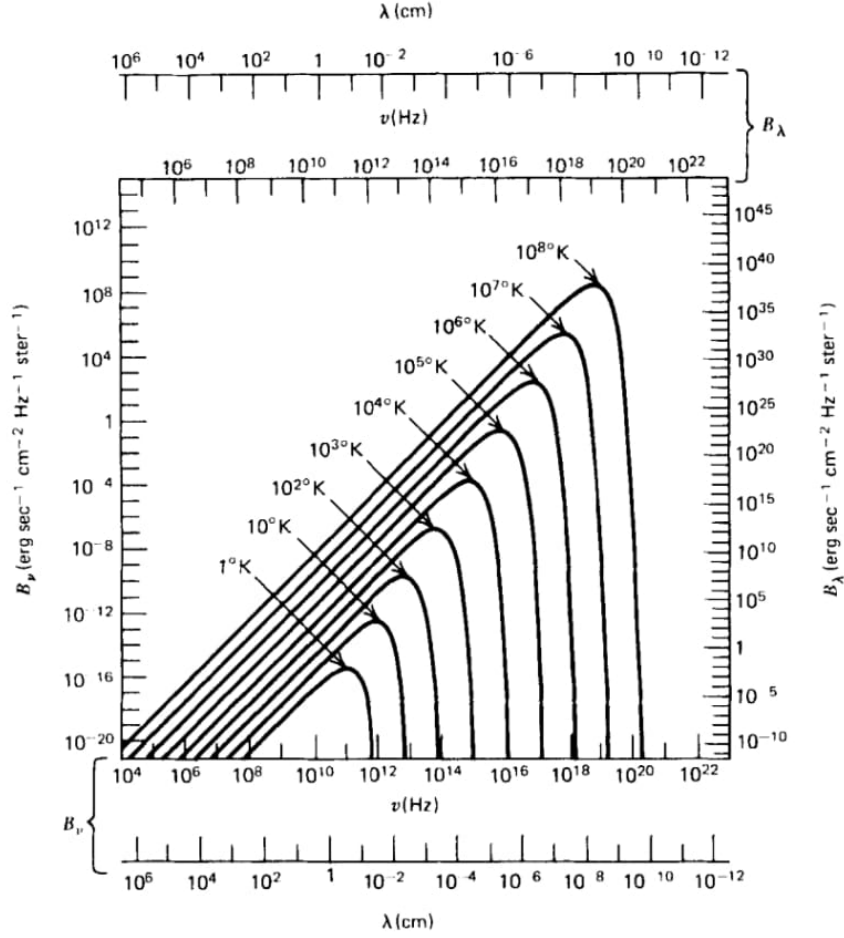


Figure B.1: Spectrum of blackbody radiation at different temperatures.

B.3 The Rosseland approximation

The radiation coming from a disc can be generally approximated as plane waves, due to its 2D-like shape. If we imagine observing a system along a direction (**s**) tilted at an angle ϑ from the direction of propagation of the waves (**z**), we can rewrite Eq. B.4 as

$$I_\nu = S_\nu - \frac{\cos \vartheta}{\alpha_\nu} \frac{dI_\nu}{dz}. \quad (\text{B.7})$$

If the emission is thermal and the region is optically thick, the intensity is very close to a Planckian, so it is almost isotropic and can therefore be expanded in series in terms of $\cos \vartheta$ as $I_\nu \approx B_\nu + \phi_\nu \cos \theta$. Substituting this expression back into the equation

APPENDIX B. GENERALITIES ON RADIATIVE TRANSPORT

above, neglecting higher-order terms and remembering that $S_\nu = B_\nu$, we find that

$$\cos \vartheta \frac{dB_\nu}{d\tau_\nu} \approx \phi_\nu \cos \vartheta, \quad (\text{B.8})$$

so

$$I_\nu \approx B_\nu - \frac{\cos \vartheta}{\alpha_\nu} \frac{dB_\nu}{dz}. \quad (\text{B.9})$$

If we now compute the monochromatic flux using Eq. B.2, we find that

$$F_\nu = -\frac{4\pi}{3\alpha_\nu} \frac{dB_\nu(T)}{dT} \frac{dT}{dz}, \quad (\text{B.10})$$

where we exploited the fact that the Planckian only depends on the temperature of the system. Computing now the total flux by integrating over all the frequencies, exploiting the Stefan-Boltzmann law and defining the *Rosseland mean absorption coefficient*

$$\frac{1}{\alpha_R} \doteq \frac{\int_0^\infty \frac{1}{\alpha_\nu} \frac{dB_\nu}{dT} d\nu}{\int_0^\infty \frac{dB_\nu}{dT} d\nu} \quad (\text{B.11})$$

we obtain

$$\mathbf{F} = -\frac{16\sigma T^3}{3\alpha_R} \frac{dT}{dz} = -\frac{c}{3\alpha_R} \frac{du}{dz}, \quad (\text{B.12})$$

where in the last equality we introduced the energy density carried by the radiation $u = \frac{4\sigma}{c} T^4$ (this relation comes from simply integrating the energy density per unit frequency, which in statistical mechanics can be written as $u_\nu = \frac{4\pi}{c} B_\nu$). From the last expression, the structure of a diffusion equation becomes evident (the energy flux is proportional to the energy gradient), implying that photons will flow towards the cooler regions.

Bibliography

- [Adeeb, 2020] Adeeb, S. (2020). Introduction to solid mechanics. <https://engcourses-uofa.ca/books/introduction-to-solid-mechanics/calculus/vector-calculus-in-cylindrical-coordinate-systems/>.
- [Armitage, 2020] Armitage, P. J. (2020). *Astrophysics of planet formation*. cambridge University Press.
- [Astropy Collaboration, 2013] Astropy Collaboration (2013). Astropy: A community python package for astronomy. *Astronomy & Astrophysics*, 558:A33.
- [Astropy Collaboration, 2018] Astropy Collaboration (2018). The astropy project: Building an open-science project and status of the v2.0 core package. *Astronomical Journal*, 156:123.
- [Cossins et al., 2009] Cossins, P, Lodato, G., and Clarke, C. J. (2009). Characterizing the gravitational instability in cooling accretion discs. *Monthly Notices of the Royal Astronomical Society*, 393(4):1157–1173.
- [Currie et al., 2023] Currie, T., Biller, B., Lagrange, A., Marois, C., Guyon, O., Nielsen, E. L., Bonnefoy, M., and De Rosa, R. J. (2023). Direct Imaging and Spectroscopy of Extrasolar Planets. In Inutsuka, S., Aikawa, Y., Muto, T., Tomida, K., and Tamura, M., editors, *Protostars and Planets VII*, volume 534 of *Astronomical Society of the Pacific Conference Series*, page 799.
- [Kataoka et al., 2014] Kataoka, A., Okuzumi, S., Tanaka, H., and Nomura, H. (2014). Opacity of fluffy dust aggregates. *Astronomy & Astrophysics*, 568:A42.
- [Kaushik et al., 2024] Kaushik, M., Mattoo, A., and Rastogi, R. (2024). Exoplanet Detection : A Detailed Analysis. *arXiv e-prints*, page arXiv:2404.09143.
- [Lodato, 2007] Lodato, G. (2007). Self-gravitating accretion discs. *La Rivista del Nuovo Cimento*, 30:293–353.
- [Lodato, 2008] Lodato, G. (2008). Classical disc physics. *New Astronomy Reviews*, 52(2):21–41. Circumstellar Disks and Planets at Very High Angular Resolution.

BIBLIOGRAPHY

- [Longarini et al., 2023] Longarini, C., Armitage, P. J., Lodato, G., Price, D. J., and Ceppi, S. (2023). The role of the drag force in the gravitational stability of dusty planet-forming disc - II. Numerical simulations. *MNRAS*, 522(4):6217–6235.
- [Paardekooper et al., 2023] Paardekooper, S., Dong, R., Duffell, P., Fung, J., Masset, F. S., Ogilvie, G., and Tanaka, H. (2023). Planet-Disk Interactions and Orbital Evolution. In Inutsuka, S., Aikawa, Y., Muto, T., Tomida, K., and Tamura, M., editors, *Protostars and Planets VII*, volume 534 of *Astronomical Society of the Pacific Conference Series*, page 685.
- [Perrin et al., 2018] Perrin, M., Duchêne, G., and Pinte, C. (2018). Welcome to MCFOST’s documentation. <https://mcfost.readthedocs.io/en/latest/index.html>.
- [Price, 2012] Price, D. J. (2012). Smoothed particle hydrodynamics and magnetohydrodynamics. *Journal of Computational Physics*, 231(3):759–794. Special Issue: Computational Plasma Physics.
- [Price et al., 2018] Price, D. J., Wurster, J., Tricco, T. S., Nixon, C., Toupin, S., Pettitt, A., Chan, C., Mentiplay, D., Laibe, G., Glover, S., Dobbs, C., Nealon, R., Liptai, D., Worpel, H., Bonnerot, C., Dipierro, G., Ballabio, G., Ragusa, E., Federrath, C., Iaconi, R., Reichardt, T., Forgan, D., Hutchison, M., Constantino, T., Ayliffe, B., Hirsh, K., and Lodato, G. (2018). Phantom: A smoothed particle hydrodynamics and magnetohydrodynamics code for astrophysics. *Publications of the Astronomical Society of Australia*, 35(e031).
- [Rice et al., 2005] Rice, W. K. M., Lodato, G., and Armitage, P. J. (2005). Investigating fragmentation conditions in self-gravitating accretion discs. *Monthly Notices of the Royal Astronomical Society: Letters*, 364(1):L56–L60.
- [Sierra and Lizano, 2020] Sierra, A. and Lizano, S. (2020). Effects of scattering, temperature gradients, and settling in protoplanetary disks. *The Astrophysical Journal*, 892:136.

Ringraziamenti

Grazie ai miei genitori, per avermi permesso di intraprendere questo percorso scelto di vivere insieme alcuni dei passaggi più importanti della nostra vita. A Pietro, Pastu e Simo, per aver mantenuto i rapporti anche dopo il liceo, nonostante non abbiano la minima idea di cosa io stia facendo. A mia sorella Manuela, per essere sempre stata un punto di riferimento con cui confrontarmi e scontrarmi. Agli zii Osvaldo e Lory, per la loro costante presenza e il sostegno incondizionato. Alle nonne Gina e Sandra, per l'affetto che mi hanno sempre dimostrato. A tutto il resto della mia famiglia, per il ruolo fondamentale che ha avuto nella mia crescita nel corso di tutti questi anni. A Nathan, per aver coltivato un rapporto davvero speciale nonostante la distanza. A Rito, Alefocchi, Bonny, GG e Cassi, per i tornei di briscola, le vacanze, le folli gite gastronomiche, le serate nei vari locali della valle; per aver condiviso esperienze uniche ed aver creato un gruppo solido e sincero. Ad Alice, per avermi spesso mostrato un diverso punto di vista e avermi fatto riflettere sull'importanza di cose che prima reputavo secondarie. A Bonny, GG e Cassi, per i tornei di briscola, le vacanze, le folli gite gastronomiche, le serate nei vari locali della valle; per aver condiviso esperienze uniche ed aver creato un gruppo solido e sincero.

RINGRAZIAMENTI

